



MASTER THESIS

Costume Design Literacy for Game Character Art Students: A Workshop-Based Investigation

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DECLARATION OF AUTHORSHIP

I, Imara van der Wel, declare that this thesis titled, "Costume Design Literacy for Game Character Art Students: A Workshop-Based Investigation" and the work presented are my own.

I confirm that:

- This work was done wholly or mainly while in candidature for a research degree at this University.
- Where any part of this thesis has previously been submitted for a degree or any other qualification at this University or any other institution, this has been clearly stated.
- Where I have consulted the published work of others, this is always clearly attributed.
- Where I have quoted from the work of others, the source is always given. With the exception of such quotations, this thesis is entirely my own work.
- I have acknowledged all main sources of help.
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Signed:



Date: 05-06-2026

This research taught me that the questions worth asking are usually the ones that sit at the edges of what you already know. Standing at the intersection of fashion and games, I found that the gap between the two disciplines was not a barrier but a starting point. I leave this study with a clearer sense of where my expertise sits and what I want to do with it.

Imara van der Wel

ABSTRACT

Academy for AI, Games and Media

Master of Game Technology

Costume Design Literacy for Game Character Art Students: A Workshop-Based Investigation

By Imara van der Wel

Game character art education provides strong technical foundations in modelling, texturing and rendering, yet costume design literacy remains absent from game art training. Studios have recognised costume expertise value through industry collaborations and adoption of pattern-based tools like Marvelous Designer, but game artists frequently lack the construction literacy to translate technical capability into deliberate narrative communication. No study had systematically identified which costume design principles are relevant and applicable for game character art students as a learning framework. This study investigated what costume design principles are relevant for game character design, and how students engage with and apply these principles. A workshop-based design was used, combining five online Guild sessions and a hands-on workshop, with a self-selected sample of game art students (n=11) across all four years of a bachelor's programme. Qualitative data was collected through naturalistic observation, verbal discussion and personal response sheets, and analysed using thematic analysis. Four themes emerged. Students engaged most productively when principles created friction with existing assumptions, resolved through their own discipline role: riggers translated fabric behaviour into deformation logic, concept artists into worldbuilding, and character artists into construction and simulation workflows. Genuinely unfamiliar materials generated broader engagement than familiar content. Semiotic understanding emerged organically through historical and material content rather than direct instruction. The principles that proved most relevant were construction logic including patterning and flat-to-3D translation, historical and material context, fabric behaviour, and semiotic thinking. These findings provide the foundational understanding needed to develop structured costume design literacy curricula within game art education.

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I wish to acknowledge how my educational background has uniquely positioned me to bridge the fields explored in this research. My training in fashion design through MBO and HBO studies, culminating in a bachelor's degree in Fashion & Textile Technologies from Amsterdam Fashion Institute (graduated February 2024), provided me with foundational knowledge in fashion history, textile science, garment construction, and pattern making. My specialization in digital fashion focused on translating traditional construction techniques into digital tools like CLO3D, while my minor in Virtual Reality introduced me to game development processes and industry terminology.

This unique intersection between fashion expertise and game production sparked the questions driving this thesis. The combination of construction literacy, material knowledge, historical context, and understanding of game development pipelines has enabled me to recognize where costume design principles naturally intersect with character production workflows, and more importantly, to identify the vocabulary translation challenges that can create barriers between these disciplines. Without this dual perspective, I might not have seen the gaps this research addresses or understood how to make costume design literacy accessible to game art students.

This research would not have taken the shape it did without the people who supported it at every stage.

My supervisors, Linda Effinger and Vlad Cenusa, each contributed something distinct and essential. Linda provided consistent weekly support throughout the entire process. Her structural guidance, careful eye for methodology, and willingness to ask the questions that kept the scope honest made this thesis more rigorous at every turn. Vlad brought an industry perspective that grounded the research in real practice, collaborated directly on teaching content within the Guild series, and asked the kinds of questions that pushed my thinking further than I would have gone alone.

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Claude (Anthropic, 2026) was used throughout this research as a working tool. During the preparation of the Guild series, it supported literature exploration and factchecking of historical content. During the writing process, it provided structural scaffolding and step-by-step task breakdowns to help manage large writing tasks. In the earlier chapters, including the literature review and introduction, it generated draft versions that the researcher then edited, restructured and rewrote to varying degrees. In the later chapters, particularly the methodology and findings, the role shifted toward feedback and structural suggestions, with the researcher writing independently and using Claude to flag clarity or structural issues. During data analysis, it asked reflective questions that helped surface connections in the data rather than generating conclusions. The researcher's engagement with AI writing support decreased deliberately over the course of the thesis as independent academic writing developed.

CONTENTS

Declaration of Authorship.....	i
Abstract	iii
Acknowledgements	iv
Contents.....	vi
List of Figures.....	viii
List of Abbreviations.....	ix
1 Introduction to the thesis topic.....	1
1.1 Background and problem context.....	1
1.2 Research questions and approach.....	1
2 Background information and theory	3
2.1 The character design challenge	3
2.2 Fashion vs costume and why the distinction matters.....	5
2.3 Educational opportunities and disciplinary gaps	6
2.4 Pattern-based tools in industry practise	8
2.5 Essential textile construction knowledge	9
2.6 Character production pipeline and costume principles application.....	10
2.7 Interdisciplinary learning.....	12
2.8 The gap.....	13
3 Methodology	15
3.1 Research design	15
3.2 Participants.....	18
3.3 Materials and tools.....	19
3.4 Procedures	21
3.5 Data analysis.....	23
4 Findings and discussion.....	26
4.1 Something I'd never seen before made me wonder what else I don't know	26
4.2 From touching fabric to rethinking design	28
4.3 New knowledge made previous work feel unfinished.....	30
4.4 The same principles, different translations: discipline role as reading lens	32
4.5 How students experienced the learning formats.....	33
4.6 Synthesis	35
5 Conclusion.....	37
6 Limitations and future directions	39
6.1 Limitations	39

6.2	Recommendations for future iterations	40
6.3	Broader research agenda	40
Appendix A: Guild series documentation		42
A.1	Session overview and structure.....	42
A.2	Introductory session content.....	42
A.3	Historical period framework.....	42
A.4	Iterative refinements.....	43
A.5	Guild series review	43
Appendix B: Hands-on workshop documentation.....		44
B.1	Workshop overview and structure.....	44
B.2	Theoretical introduction content	44
B.3	Station descriptions	44
B.4	Workshop end	45
B.5	Personal response sheet questions	45
Appendix C: Supporting quotes by theme.....		46
C.1	Findings & discussion 4.1	46
C.2	Findings & discussion 4.2	47
C.3	Findings & discussion 4.3	48
C.4	Findings & discussion 4.4	49
C.5	Findings & discussion 4.5	50
Bibliography		53

LIST OF FIGURES

Figure 1 - Ellie Williams, The Last of Us.....	3
Figure 2 - Jill Valentine, Resident Evil 3	3
Figure 3 - Character production pipeline	11
Figure 4 - The Experiential Learning Model.....	17
Figure 5 - Four themes emerging from data analysis.....	26

LIST OF ABBREVIATIONS

Barding	Decorative and protective covering fitted to a horse or mount, typically made from fabric, leather or metal. In costume design, barding follows the same construction logic as human armour and clothing, adapted for animal anatomy.
Basic block	A foundational pattern template drafted to a set of body measurements, with no style details added. It serves as the starting point from which all other garment patterns are developed.
Bias	The diagonal direction across a woven fabric, running at 45 degrees to the warp and weft. Fabric cut on the bias has more stretch and drape than fabric cut on the straight grain, producing a fluid silhouette.
Calico	An undyed, unfinished plain-weave cotton fabric. Commonly used in garment construction for making toiles (test versions of a garment) before cutting into final fabric.
Character production pipeline	The sequential workflow through which a game character moves from initial concept to final in-game implementation. Stages typically include concepting, block-out modelling, high poly sculpt, retopology, UV mapping, baking, texturing, rigging and implementation.
Cloth simulation	A digital process in which software calculates how a fabric behaves physically, including how it folds, drapes and responds to gravity and movement. Used in game development to generate realistic garment geometry.
CL03D / Marvelous Designer	Pattern-based 3D garment simulation software used across the fashion and game industries. Garments are constructed from flat pattern pieces that are sewn together and simulated to produce three-dimensional cloth geometry.
Corsetry / boning	Corsetry refers to structured undergarments designed to shape the torso. Boning refers to the rigid strips inserted into a corset to provide structure, historically made from materials such as whalebone (baleen) and later steel or synthetic alternatives.
Costume design	The practice of designing clothing that serves a narrative purpose, communicating character identity, social status, cultural background and story context. Distinct from fashion design in that it prioritises fictional coherence over trends or self-expression (Salomaa, 2021).
Draping	A garment construction method in which fabric is arranged directly on a dress form or body to create shape and volume, rather than working from a flat pattern. Also refers to the way fabric falls and folds under gravity.
Fashion design	The practice of designing clothing driven by aesthetic trends, self-expression and commercial appeal. Distinguished from costume design by its focus on the garment as a cultural or commercial object rather than a narrative one (Salomaa, 2021).

Flat-to-3D translation	The process by which a flat, two-dimensional pattern piece is sewn or assembled into a three-dimensional garment form. Understanding this translation is foundational to both physical garment construction and digital simulation workflows.
High poly / low poly	High poly refers to a 3D model with a large number of polygons, used during sculpting to capture fine surface detail. Low poly refers to a simplified version of the same model optimised for use in a game engine. Detail from the high poly is transferred to the low poly through a process called baking.
Pattern making	The process of creating the flat template pieces from which a garment is cut and assembled. Pattern pieces define the shape, fit and construction logic of a finished garment.
Rigging	The process of creating a digital skeleton and control system for a 3D character model, enabling it to be posed and animated. Costume elements such as layered clothing, accessories and long panels affect rigging complexity.
Semiotics / vestimentary code	Semiotics is the study of signs and symbols as systems of communication. Applied to clothing, the vestimentary code refers to the way garments function as a coded visual language, communicating information about social status, identity, cultural belonging and narrative context to viewers, often below conscious awareness (Honeycutt, 2021).
Silhouette	The overall outline or shape of a garment or costume as seen against the body. Silhouette is one of the primary visual signals a costume sends, communicating period, status and character type before any detail is visible.
Texturing	The process of applying surface detail to a 3D model using digital image maps. In character art, texturing determines the visual appearance of materials including colour, surface roughness, reflectivity and fine surface detail.
UV mapping	The process of unwrapping a 3D model's surface into a flat 2D layout so that texture maps can be applied accurately. UV maps determine how a 2D texture image aligns with the three-dimensional geometry of a model.
Warp and weft	The two sets of threads that form a woven fabric. The warp runs lengthwise along the fabric and is held under tension on the loom. The weft runs crosswise, interlacing with the warp to create the woven structure. The relationship between warp and weft determines the fabric's weave type, behaviour and properties.

For everyone who ever wondered why the costume in-game looked wrong.

1 INTRODUCTION TO THE THESIS TOPIC

Games face a familiar challenge that film and theatre navigated decades ago: how to integrate costume knowledge meaningfully into creative production. This chapter establishes the problem context driving this research, outlines the gap between technical capability and design reasoning in current game art education, and introduces the research questions and investigative approach that structure the study.

1.1 BACKGROUND AND PROBLEM CONTEXT

Film and theatre confronted a familiar challenge decades ago: how to systematically integrate costume knowledge into creative production. Film evolved from casual wardrobe departments to professionalized costume design by the 1920s, establishing costume as a character development tool rather than decorative afterthought (Stutesman, n.d.). Theatre used semiotic frameworks to use construction choices to encode character information (Jablon-Roberts & Sanders, 2018). Now games face a similar integration challenge, but with a significant difference: pattern-based tools like Marvelous Designer exist and studios recognize costume expertise value through industry collaborations, yet integration of costume design literacy into game art education remains unexplored.

Game art curricula provides strong technical foundations in modelling, texturing and rendering (Czauderna, 2018). Salomaa discusses that the industry has adopted pattern-based tools like Marvelous Designer as standard practice and studios recognize costume design value enough to hire external consultants. But a gap persists between technical execution and design reasoning. Even when studios collaborate with costume professionals, different vocabularies and workflows create communication barriers (Makryniotis, 2018; Salomaa, 2019). Artists can operate the tools but often lack the construction literacy that transforms technical capability into deliberate narrative communication. So, education provides the technical foundation, industry provides the tools and validates expertise value, and yet the pedagogical bridge of integrating costume design literacy into game art training remains undeveloped.

1.2 RESEARCH QUESTIONS AND APPROACH

While industry collaborations prove costume expertise enhances character design (Cork, 2015) and pattern-based tools demonstrate construction thinking's value, fundamental questions remain unexplored. Game design doesn't need costume design's full

breadth, but it still needs the principles that enable deliberate narrative communication through clothing. But which principles are those? Historical context matters for believability (Jablon-Roberts & Sanders, 2018; Lyons, 2020), textile and construction logic informs how garments behave (Adams, 2023; Honeycutt, 2021), and semiotics reveals how clothing communicates character information (Honeycutt, 2021; Tran, 2019). Yet determining what knowledge specifically serves game contexts, and how to integrate that literacy into existing technical training without overwhelming students, remains uncharted territory.

This research investigates what costume design principles are essential for narrative driven game character design, and how they can be effectively integrated into game art education through interactive workshop methodologies. The study unfolds in two phases. Phase one will synthesize existing literature and analyse industry examples to identify foundational principles like historical context, dress semiotics and construction thinking. And phase two is to design and test interactive workshop formats that teach these principles through hands-on material engagement which includes material engagement, iterative experimentation and connecting construction choices to narrative purpose.

This study finds that students translate costume design principles through the lens of their own discipline and discipline role, and that this translation generates new ideas, shifts their design thinking and results in more deliberate creative decisions. The research contributes foundational understanding of which costume design principles resonate with game art students and how interdisciplinary knowledge can be introduced through hands-on workshop methodologies, providing a basis for future development of structured teaching approaches in game art education.

2 BACKGROUND INFORMATION AND THEORY

This chapter surveys the existing literature and industry practice that frames the research. It examines how costume design functions as a narrative communication system, why the distinction between fashion and costume design matters for game contexts, where educational and disciplinary gaps currently sit, and how pattern-based tools and interdisciplinary learning theory together point toward the gap this study addresses.

2.1 THE CHARACTER DESIGN CHALLENGE

Character design in games operates as a complex visual communication system that extends far beyond aesthetic appeal. Costume design creates symbolic communication, with Honeycutt's analysis showing how exploring 'the linguistic essence of each garment, each costume, and each colour choice' enables meaningful symbolic and cultural conclusions (Honeycutt, 2021, p. xii). The difference between successful and unsuccessful costume implementation in games shows varying levels of symbolic literacy application in character design practice. A successful example of this is *The Last of Us*, which achieves narrative depth partly through costume choices that communicate character arc, history and world-building (Neonakis, 2022), while less successful designs often feature generic or narratively disconnected clothing that fails to enhance storytelling, like the outfit of Jill Valentine from *Resident Evil 3* whose clothing fails to suggest she is either the protagonist or a law enforcement officer. (Tran, 2019)



Figure 1 - Ellie Williams, *The Last of Us*
(Reddit user: FreeWear2215, 2025)



Figure 2 - Jill Valentine, *Resident Evil 3*
(vhv.rs, 2019)

The distinction between successful and unsuccessful costume implementation extends beyond photorealistic games into stylized visual styles. Salomaa's research emphasizes

that costume design principles apply across visual approaches, addressing both photorealistic games and those representing stylized realism (Salomaa, 2018). While stylized art deliberately departs from photorealism through exaggerated proportions, simplified forms, or vibrant colour palettes, believability remains essential for narrative coherence. Material choices significantly influence believability regardless of visual style, as texturing decisions must align with the game's internal logic even when that logic differs from physical reality (Shahbazi, 2025).

Real-world garment construction knowledge informs deliberate stylization choices rather than constraining artistic expression. Games employing stylized realism combine realistic rendering techniques with exaggerated character designs (Tereshchuk, 2025). *It Takes Two* (Hazelight Studios, 2021) exemplifies this approach, where Hazelight Studios pursued "a stylized approach but without losing the realism, using tiling textures that helped sell the various materials in the world" (Wade, 2021). Character costumes demonstrate construction literacy through recognizable textile details, maintaining "a noticeable difference in the textures" despite stylization (Hawes, 2021), demonstrating how construction knowledge enables informed decisions about when to follow or deliberately break physical garment logic for stylistic effect.

Understanding costume design's communicative function requires recognizing clothing as a coded system. As Honeycutt demonstrates through Barthes' vestimentary theory, "clothing did not evolve as a coded system like that of grammar, the quest for its meanings lies in its "functions, distinctions, oppositions, and congruences" (Barthes, 1983, pp. 293-295, as cited in Honeycutt, 2021, p. 11). Recognizing clothing as a coded system distinguishes costume design literacy from general visual skill, which relies on cultural and domain-specific knowledge to decode symbolic meaning (Honeycutt, 2021). Tran articulates this as fashion being "its own complex language. Codified, meaningful, and symbolic, with different contexts and moods" (Tran, 2019), emphasizing that understanding this visual language enables designers to increase understanding and send signals that deepen world-building.

Industry collaboration patterns demonstrate this expertise gap. Cork's interview with Eidos Montreal shows that they explicitly sought costume professionals despite employing skilled character artists (Cork, 2015), while Kucharski identifies costume design as an "upswing" area where professional expertise creates measurable improvements in character believability (Kucharski, n.d.), suggesting existing skill sets benefit from specialized knowledge. Lyons's work positions costume design knowledge as strategic problem-solving capability that complements existing game art skills (Lyons, 2020).

The historical development of costume design theory provides essential context for understanding contemporary challenges. Stutesman's documentation of early French film shows costume designers who treated characters "as whole entities", requiring garments that reflected personality, occupation, cultural background and narrative function simultaneously (Stutesman, n.d.). Honeycutt's research demonstrates how costume elements like colour, cut, and composition create semantic meaning below conscious viewer awareness (Honeycutt, 2021), while Neonakis discusses "costuming a character's narrative arc" to show how clothing choices advance storytelling objectives (Neonakis, 2022). These principles appear in *Horizon Zero Dawn*, where post-apocalyptic costume design balances fantasy elements with believable material culture, creating visual authenticity that supports world-building objectives (Davies, 2017). These examples demonstrate costume design as systematic narrative tool where construction knowledge, cultural literacy, and visual storytelling intersect; a framework that transforms technical execution into purposeful character communication.

The tension between believability and realism in digital garments represents a critical challenge in current game character design. Tran emphasizes that fashion as a "codified, meaningful, and symbolic" language can "send different signals that deepen the understanding of your world" (Tran, 2019). However, Tepe et al. identifies a tension within fashion-game industry collaborations: even when fashion-literate designers are integrated into game development, the resulting garments "often result in garment designs that resemble physical equivalents rather than constituting innovative experiences" (Tepe & Koohnavard, 2022, p. 1). These perspectives suggest that believability emerges not from literal reproduction, but from informed manipulation of textile principles that align with narrative requirements.

2.2 FASHION VS COSTUME AND WHY THE DISTINCTION MATTERS

Costume design in games prioritizes narrative context over trends, creating character-specific clothing that communicates story information through visual language. Naughty Dog's approach to costume design in *The Last of Us* embodies this principle, where each character's weathered, scavenged garments reflect twenty years of post-apocalyptic survival while maintaining individual personality markers that distinguish characters within shared environmental constraints as described by Neonakis (Neonakis, 2022). Guerrilla Games' *Horizon Zero Dawn* provides another clear example of costume design methodology throughout the design process, as documented in *The Art of Horizon Zero Dawn*. Designers conducted extensive research into geographical determinism, using environmental factors to inform how tribes would develop: their social hierarchies, cultural expression patterns, and technological capabilities. These research-driven decisions directly shaped material selection and garment

construction choices (Davies, 2017). Salomaa discusses that these examples show how costume design prioritizes narrative coherence over aesthetic appeal, where every visual element must justify its existence within the fictional context rather than serving external aesthetic or commercial objectives (Salomaa, 2021).

In contrast, fashion-oriented games like Fortnite, Pokémon GO and Animal Crossing use fundamentally different design principles, which prioritize self-expression and social interaction over narrative coherence. Unlike narrative driven games, games that focus more on self-expression embrace individual identity and modern trends designed for mass appeal, creating customization ecosystems where players construct and communicate personal identity through clothing choices that transcend the game's fictional boundaries (Salomaa, 2021). The distinction becomes especially clear in collaborations, where fashion brands like Balenciaga partner with games to extend their cultural reach through digital twins of physical collections, treating game spaces as alternative fashion platforms rather than narrative environments which would require context-specific design integration (Epic Games, 2021).

While both fashion and costume design share foundational understanding of patternmaking, draping, and fabric behaviour, their application changes based on design objectives. Fashion design demands literal construction accuracy with recognizable closures, seams, and structural details that mirror real-world garments, whereas costume design permits creative reinterpretation of construction logic, as long as the results maintain narrative believability within the fictional world (Salomaa, 2021). This research focuses specifically on costume design principles; the narrative-driven, character-specific approach, as this represents the primary professional context for game character designers. However, which specific textile construction principles prove most relevant for game contexts, and how designers might distinguish narrative-essential knowledge from fashion design's broader domain, remains unexplored in existing literature. These questions connect directly to documented gaps between technical capability and design reasoning in current game art education.

2.3 EDUCATIONAL OPPORTUNITIES AND DISCIPLINARY GAPS

Current game art curricula provide strong foundational skills in technical modelling and visual design, yet present opportunities for deeper integration of interdisciplinary knowledge that could enhance character design outcomes. Czauderna's comparative study shows that current programs successfully teach modelling, texturing, and rendering capabilities but have room for expansion in interdisciplinary knowledge that could inform design choices (Czauderna, 2018). This educational foundation intersects with growing industry recognition of specialized knowledge value, as demonstrated when Eidos Montreal's

executive art director acknowledged that game developers "have to create entire worlds... but we're none of those things" and emphasized the benefits of consulting "real people" with domain expertise (Cork, 2015). This suggests that education stands to benefit from including knowledge fields that complement existing technical instruction.

Makryniotis's analysis identifies gaps between fashion professionals and game artists as stemming from different vocabularies and methodological approaches rather than incompatible skill sets (Makryniotis, 2018). Fashion professionals discuss garment construction using terms like "bias cut" or "dart placement" while game artists describe the same visual outcomes through rendering-focused language like "fabric flow" or "form-fitting silhouette." These differing frameworks reflect distinct ways of understanding the same information: fashion professionals decode why a garment behaves as it does through construction logic, while game artists observe what it looks like through visual analysis. Developing shared terminology could enable game artists to access construction-based reasoning that currently remains specialized knowledge.

Robinson's analysis of interactive learning methodologies suggests practical frameworks for developing these bridging capabilities, indicating that educational approaches could address translation challenges between fashion understanding and technical implementation (Robinson & Alvarez, 2024). These findings together identify a clear gap: while industry recognizes costume design value and collaboration challenges are documented, the pedagogical mechanisms for integrating construction knowledge into game art education remain largely undeveloped. This gap becomes particularly significant given widespread industry adoption of pattern-based tools that require construction literacy for effective application.

Relying solely on external costume expertise presents several limitations. Most game studios, particularly independent developers, lack budgets to hire specialized costume consultants for every project, yet these teams still use pattern-based tools like Marvelous Designer that assume construction knowledge (Zealous XR, 2025). Even when studios can afford external expertise, documented communication barriers between fashion professionals and game artists (Makryniotis, 2018; Salomaa, 2019) suggest both parties benefit when game artists possess baseline costume design literacy that enables productive collaboration rather than requiring complete translation of specialized vocabulary. Costume decisions occur throughout the character production pipeline from initial concepting through final implementation (as discussed in Section 2.6), not at a single consultation point, meaning artists

make construction-related choices daily that benefit from integrated knowledge rather than external validation.

2.4 PATTERN-BASED TOOLS IN INDUSTRY PRACTISE

Marvelous Designer (CLO Virtual Fashion, n.d.) has established itself as industry standard for realistic 3D cloth modelling in game development, described as "the go-to tool for character design, animation, and visual effects studios" with adoption across major entertainment companies including HBO, Walt Disney, EA Games, and Gameloft (Zealous XR, 2025). The software's Academy Award-winning technology (CLO Virtual Fashion, 2024) distinguishes itself from general-purpose 3D software like Maya (Autodesk, n.d.) or Blender (Blender Foundation, n.d.) through its core focus: simulating real-world pattern construction and physics-based fabric behaviour rather than requiring manual sculpting of cloth appearance (Zealous XR, 2025). This allows artists to work with garment construction logic rather than purely visual approximation. Major studios have integrated this pattern-based approach into their core pipelines for over a decade, with *The Witcher 3* and *Cyberpunk 2077* demonstrating advanced applications where artists used construction approaches that transcend physical garment limitations within digital creative freedom (Marvelous Designer, 2017a, 2017b). This widespread adoption demonstrates that realistic character clothing production has shifted fundamentally toward garment construction thinking as standard practice rather than experimental technique.

Multiple garment creation methods exist within game development pipelines, each with its unique advantages. 3D scanning provides unparalleled detail capture and reduces production time for realistic assets (InspirationTuts, 2022). Sculpting software like ZBrush (Maxon, n.d.) creates highly detailed fabric through manual manipulation and frequently pairs with simulation tools to add fine detail after initial cloth simulation (Gear Solid, 2024; InspirationTuts, 2025). General-purpose software like Blender and Maya offer cloth simulation integrated within broader toolsets, making them cost-effective for projects where garment creation represents one component among many. While technical tools like Houdini excel in complex cloth interactions with environmental elements, though they require substantial technical expertise (Gear Solid, 2024). Despite the range of available tools, pattern-based approaches remain the industry standard for character costume work precisely because they require and reinforce construction thinking, the literacy this study addresses.

Pattern-based tools like Marvelous Designer have become industry standard for character costume creation because they mimic real-world garment construction processes, allowing artists to work with flat fabric patterns that behave as physical garments would (Gear

Solid, 2024). However, industry practitioners reveal that pattern-based software alone does not constitute complete workflows. Pedram Karimfazli, lead character artist at EA Criterion, describes the established pipeline: "Once we have the base, we bring this into ZBrush where we clean it up and add more detail using classic sculpting techniques, custom brushes, and alphas" (Karimfazli, 2019). This MD-to-ZBrush workflow has become standard practice where simulation provides construction-accurate base geometry that artists then refine with sculpted detail (Gear Solid, 2024; Karimfazli, 2019). Pattern-based approaches present trade-offs: dense meshes require retopology for animation optimization, the tools demand sewing knowledge or willingness to learn construction principles, and high-polygon garments require substantial computing power (Gear Solid, 2024). Despite these constraints, pattern-based tools' ability to produce realistic garment behaviour efficiently has established them as the go-to solution for character costume work.

This tension between tool capability and artist knowledge becomes visible when studios seek external expertise. Cork's documentation of the Deus Ex: Mankind Divided collaboration reveals how Eidos Montreal's development team explicitly acknowledged lacking specialized fashion expertise despite creating "entire worlds," leading them to partner with professional fashion designers from Acronym who approached character design through authentic garment construction thinking, demonstrably improving character design through "research over assumption" approaches (Cork, 2015). Yet Salomaa's work for Remedy Entertainment's Control, shows that even when costume expertise is available, effective collaboration requires bridging different professional vocabularies, workflows, and design approaches between fashion professionals and game artists (Salomaa, 2019). Together, these patterns suggest that while pattern-based tools have become standard, construction knowledge determines effective application. This raises questions about what construction knowledge is essential for artists to leverage these tools effectively, and how that knowledge might be integrated into game art education.

2.5 ESSENTIAL TEXTILE CONSTRUCTION KNOWLEDGE

Understanding textile behaviour and construction techniques serves character storytelling through observable material qualities that communicate narrative information. Fabric conveys character details that viewers intuitively decode: while detailed semiotic analysis requires "meticulous extraction," viewers' brains process clothing codes "almost automatically and at lightning speed" during typical film viewing (Honeycutt, 2021, p. 92). Yet translating intuitive observations into deliberate design choices requires understanding the construction logic behind visual outcomes. Lyons's approach to teaching costume history for

game art students demonstrates this through drawing-based practice, arguing that kinetic to visual input is one of the most effective means of retaining information and that physically drawing costumes from each historical period stores that knowledge in long-term memory (Lyons, 2020). This material literacy enables more deliberate design decisions, grounding choices in the actual behaviour of textiles rather than assumption.

Historical and cultural context transforms period-accurate costume details into visual communication tools that convey character information to audiences. Costume designers act as semioticians, in this case, they interpret linguistic signs from production texts and translate them into visual signs through costume elements calculated to communicate character traits and time period. As Broadway designer Tracy Christensen states, "Costumes contribute major visual information to the audience, and 'speak' before the actor speaks." Through strategic selection of construction techniques, silhouettes, and fabrics, designers communicate social status and wealth through visual cues that audiences interpret using shared cultural codes, often subconsciously (Jablon-Roberts & Sanders, 2018). Lyons's pedagogical approach demonstrates this application practically, teaching students how period-specific textile production methods, dye availability, and construction technologies inform material selection that maintains believability while serving narrative functions (Lyons, 2020). This historical lens proves particularly valuable for fantasy and fictional environments common in game development, where understanding how real-world cultures used textiles for status display, cultural identity, and practical functionality informs believable fictional costume systems.'

The textile and construction knowledge identified here: material behaviour, historical context, and construction logic, informed the selection of principles that would later form the foundation of the workshop design, as discussed in the methodology.

2.6 CHARACTER PRODUCTION PIPELINE AND COSTUME PRINCIPLES APPLICATION

Character production pipelines progress through distinct phases where different types of costume knowledge become relevant. Industry practitioners document character production workflows as progressing from conceiving through 3D modelling to final implementation (Figure 3) (Chavez, 2025; Pikat, 2024; Proko & Evstratiy, 2024; Proko & Zhang, 2021; Riot Games, 2018). This pipeline structure reflects current industry practice and aligns with game art curricula taught in Year 2 Visual Arts courses at Breda University of Applied Sciences (V. Cenus, personal communication, 2025).

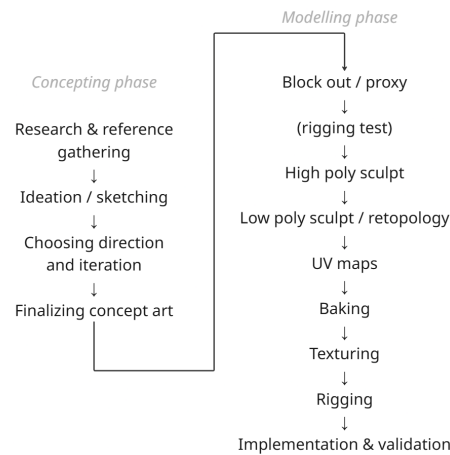


Figure 3 - Character production pipeline

Historical and cultural knowledge informs the conceiving phase, particularly during research and reference gathering. Understanding period-accurate details, material availability, and cultural dress codes shapes believable character development (Honeycutt, 2021; Jablon-Roberts & Sanders, 2018; Lyons, 2020). Lyons demonstrates how historical textile production methods and dye availability inform material selection for fantasy and fictional environments, where real-world cultural textile use provides foundation for believable fictional costume systems (Lyons, 2020). Social status and cultural dress codes translate character traits into visual communication through costume elements that audiences interpret using shared cultural codes (Jablon-Roberts & Sanders, 2018).

Construction and pattern knowledge becomes critical during 3D modelling when artists work with pattern-based simulation software. Pattern-based tools like Marvelous Designer translate construction logic from physical garment production into digital workflows, requiring understanding of how pattern pieces connect, how garment assembly creates three-dimensional forms, and how seam placement affects silhouette. Lyons's draping chapter demonstrates that understanding how weave direction, fabric weight and tensile strength affect fold behaviour gives artists a reliable foundation for working with garments convincingly, whether in illustration or digital simulation (Lyons, 2020). This construction thinking extends into sculpting refinement, where artists must understand layering logic: how undergarments sit beneath outer layers, how closures function, and how volume distributes across the body.

Textile and material knowledge applies throughout conceiving and 3D modelling phases. Understanding fibre types and fabric construction methods informs material selection that serves narrative purposes. Natural fibres in historical settings versus synthetic materials in futuristic contexts enhance world-building authenticity, as demonstrated in *The Witcher 3* versus *Cyberpunk 2077* (Marvelous Designer, 2017a, 2017b). Fabric properties like

weight, stiffness, and drape behaviour affect how garments fall and move. This knowledge becomes technical during texturing, where artists benefit from understanding how fibre content, weave structure, and surface treatments create visual qualities they replicate digitally.

Narrative and semiotic knowledge spans the entire pipeline but proves particularly critical during conceiving and believability assessment. Honeycutt demonstrates that clothing functions as a coded system where viewers decode semantic meaning automatically, processing costume information rapidly (Honeycutt, 2021). Tran describes fashion as "its own complex language" that designers use to "send different signals that deepen the understanding of your world" (Tran, 2019). This requires cultural and domain-specific knowledge, as costume choices communicate character arc, social status, and personality through visual codes. Neonakis discusses "costuming a character's narrative arc" to show how clothing choices advance storytelling objectives (Neonakis, 2022).

Believability assessment requires evaluating whether construction choices, material selections, and historical details align with narrative context and world-building requirements (Davies, 2017; Honeycutt, 2021; Tran, 2019). Davies documents how Horizon Zero Dawn balanced fantasy elements with believable material culture through research-driven decisions about environmental factors and how contextual constraints shaped material selection and construction (Davies, 2017). This assessment spans the pipeline from initial concept validation through final implementation.

While these costume design principles appear throughout costume design literature, their specific application within game character production pipelines and pedagogical integration into game art education remain unexplored.

2.7 INTERDISCIPLINARY LEARNING

The film industry example demonstrates successful integration of costume design specialization into creative production pipelines. Stutesman's historical analysis documents how early cinema evolved from informal "wardrobe" approaches to professionalized "studio design" departments by the 1920s, with designers like Clare West establishing costume as systematic character development tool rather than decorative afterthought (Stutesman, n.d.). Film's trajectory from casual wardrobe to professionalized costume departments shows a broader principle about visual storytelling mediums: recognizing clothing as narrative communication rather than decoration drives the integration of specialized knowledge.

Interactive workshop methodologies provide frameworks for developing practical skills through experiential learning. Robinson's analysis of design-focused teaching

approaches and Paine & McMahon's examination of workshop models reveal that hands-on engagement with materials and iterative experimentation produce deeper understanding than lecture-based knowledge transfer alone (Paine & McMahon, 2023; Robinson & Alvarez, 2024). These pedagogical approaches align with Lyons's observation that kinetic-to-visual learning creates stronger retention, the principle that "if you draw it, you remember it" extends to physical material engagement where handling fabrics, observing drape behaviour, and manipulating construction elements builds intuitive understanding (Lyons, 2020). Similarly, Flitton Adams's pedagogical framework provides concrete application of these principles, demonstrating that students who physically engage with textiles through handling different fabric weights and sketching construction details retain principles more effectively than those receiving passive instruction (Adams, 2023). Workshop approaches that make explicit how clothing communicates meaning below conscious awareness: through observation exercises, material handling, and construction analysis, can develop students' ability to deliberately manipulate these visual codes for character storytelling.

Taken together, these pedagogical frameworks suggest that game development faces a parallel challenge to what film and theatre navigated decades earlier: integrating specialized costume knowledge into creative production pipelines. Film's evolution from casual wardrobe to professionalized costume departments (Stutesman, n.d.) established costume as a systematic character development tool, while theatre formalized the semiotic framework through which construction choices encode character information (Jablon-Roberts & Sanders, 2018). Interactive workshop methodologies offer games a similar integration path. When positioned throughout the character development pipeline instead of isolated technical training, this progression from observation to material engagement to digital execution enables students to approach tools like Marvelous Designer with the design thinking that informs costume design practice.

2.8 THE GAP

The converging lines of film, theatre, and pedagogical evidence reveal what remains unknown in game character design education. Film and theatre integrated costume knowledge into their creative production pipelines decades ago, establishing systematic approaches where costume thinking became foundational to character development (Jablon-Roberts & Sanders, 2018; Stutesman, n.d.). Games recognize costume expertise value through industry collaborations and widespread adoption of pattern-based tools like Marvelous Designer (Cork, 2015; Marvelous Designer, 2017a, 2017b; Zealous XR, 2025) yet significant gaps persist.

Communication barriers exist even when costume professionals collaborate with game studios (Makryniotis, 2018; Salomaa, 2019). Pattern-based tools require construction literacy to achieve narrative depth beyond technical execution, as Karimfazli's MD-to-ZBrush workflow demonstrates (Karimfazli, 2019). Workshop methodologies prove effective for interdisciplinary knowledge transfer (Paine & McMahon, 2023; Robinson & Alvarez, 2024), yet remain untested for this specific application. Most critically, costume design encompasses extensive specialized knowledge, but which specific principles prove essential for game character design contexts remains unexplored.

While existing research identifies the value of costume expertise in games and documents communication barriers between disciplines, no study has systematically identified which costume design principles are relevant and applicable specifically for game character art students as a learning framework, and no pedagogical methods have been developed to integrate that literacy into game art training. This gap becomes particularly significant given that game art curricula already demand substantial technical mastery and the goal is not to create costume design specialists, but equipping character artists with literacy that enhances design decision-making and enables effective collaboration with costume professionals when studios seek specialized expertise.

Because no prior research has identified which costume principles transfer effectively to game contexts or documented how game art students engage with interdisciplinary textile knowledge, establishing this foundational understanding requires exploratory investigation. This research cannot begin with hypothesis-testing about effectiveness of predetermined principles because the principles themselves have not yet been identified and validated for game character contexts. Instead, this study must start by observing what happens when students encounter costume design concepts: which principles generate immediate recognition of applicability to their character work, which concepts require translation between disciplines, and how students articulate connections between construction literacy and their existing technical pipelines.

This research asks: What costume design principles are relevant for game character design, and how do students engage with and apply these principles in workshop contexts? The question focusses on two linked problems: figuring out *what* costume knowledge matters for games requires watching *how* students work with that knowledge, while understanding student engagement requires introducing diverse costume principles to see which ones students find most useful. This exploratory approach creates the foundation for future research to develop structured teaching methods based on the principles identified here.

3 METHODOLOGY

This chapter describes how the research was conducted. It covers the overall research design and the reasoning behind methodological choices, who the participants were and how they were recruited, what materials and tools were used across both delivery formats, how sessions were run in practice, and how the collected data was analysed. Where decisions were made or adjusted during the process, the reasoning behind those adjustments is documented here.

3.1 RESEARCH DESIGN

This study uses two formats of delivery: Guilds and a workshop. At BUAs, each Friday is reserved for group learning, where teachers or students can run sessions about a specific topic. For this research, the costume design Guild was created to offer this topic as a group learning activity. The costume design Guild consisted of online 1-hour long sessions of theoretical content delivery using costume principles applied to historical periods to explain how each principle manifested throughout time. Online-only delivery was chosen over hybrid based on testrun findings and student schedule analysis, as it maximised accessibility without additional preparation burden.

A workshop-based design research approach was selected to document how game art students engage with and apply costume design principles relevant to game character education, that were identified through literature synthesis and analysis of the character production pipeline prior to the data collection.

A workshop-based intervention with naturalistic observation was selected because this kind of research treats the teaching environment as research sites where curriculum development and empirical investigation occur simultaneously (Paine & McMahon, 2023). This approach was chosen over controlled experimental designs because the research addressed two interconnected gaps: which costume principles matter for game contexts (a knowledge gap) and how students could engage with hands-on activities (a pedagogical gap).

The original intent followed Robson and McCartan's triangulation strategy combining quantitative and qualitative data (Robson & McCartan, 2016, p. 179). Including pre/post assessment questions to quantify knowledge change. However, as data collection progressed it became clear that the research question: how students engage with and apply costume principles, is fundamentally a qualitative question. Quantitative confidence ratings could indicate that students felt more knowledgeable but could not capture which concepts resonated, how students translated principles into their own discipline, or what connections emerged spontaneously during activities. The shift to fully qualitative analysis was therefore not

a limitation but a realignment with what the research question actually required. Following Patton's argument that convergence itself constitutes evidence of credibility within a purely qualitative paradigm (Patton, 2002, pp. 559-560), three complementary approaches were used: observation documenting behavioural engagement during activities, during-activity verbal data manifesting in spontaneous comments, connections and reactions while theoretical content was being delivered or hands-on work was happening, and post-activity or post-topic group discussion with structured reflective talk.

The two formats drew on different theoretical foundations for their discussion structures. In the Guilds, structured prompts were placed at strategic points throughout each session. Prompts placed before a new topic functioned as prior knowledge activation, orientating students toward incoming content by surfacing what they already knew or assumed. Prompts placed after a topic or at the end of the full session functioned as focused discussion, inviting students to articulate connections between the costume principles just covered and their own character design practice. Both types kept discussion directed and researcher-guided, with student responses forming the primary verbal data source for the Guild sessions.

The hands-on workshop operated differently. Without structured prompts guiding verbal responses, student reactions emerged spontaneously during activities as comments, questions, connections and curiosities that arose naturally from physical engagement with materials. Following Franz's observation that unstructured group discussion can reveal valuable insights into areas not previously considered by the researcher (Franz, 2014, p. 1383), these spontaneous verbal responses were treated as naturalistic data revealing which aspects of costume knowledge students found immediately relevant without researcher direction.

Together, the Guilds and the hands-on workshop formed a single integrated learning experience, designed so that each format completes what the other could not. The Guilds offered theoretical grounding and structured reflection across historical and cultural contexts, while the workshop provided the physical, tactile engagement that online delivery couldn't offer. This complimentary design was informed by Kolb's (1984) experiential learning theory, following the Lewinian Experiential Learning Model. (*figure 4*) Kolb positions learning as a cyclical process integrating concrete experience, reflective observation, abstract conceptualisation and active experimentation (p. 21). The workshop engaged all four stages, moving through concrete experience via physical material engagement, reflective observation through reasoning about material behaviour, abstract conceptualisation through theoretical framing, and active experimentation through brainstorming design applications, while the

Guilds operated the cycle partially, covering abstract conceptualisation and active experimentation without the concrete tactile element the workshop provided.

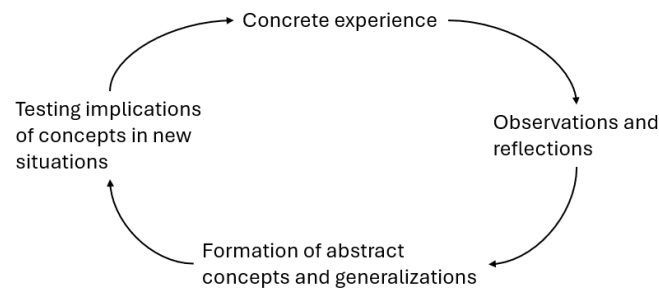


Figure 4 - The Experiential Learning Model

The research unfolded across three phases with iterative development. Phase one involved preliminary literature synthesis to identify foundational principles shared between fashion and costume design (pattern construction, fabric behaviour, draping techniques) while recognizing that determining which principles are most relevant for game character design required direct observation of student engagement. Phase two tested workshop methodology through three sessions teaching fashion and digital garment creation foundations: garment production pipelines, textile properties and construction methods, and digital fashion tools. These testrun workshops served dual purposes of refining facilitation techniques and data collection procedures while observing how students interacted with hands-on construction activities to inform subsequent content refinement. Phase three consisted of the main data collection across two complementary formats: five Guild sessions covering costume principles applied to historical periods, and a hands-on workshop exploring textile, fabric behaviour and patterning through physical activities. Data was gathered through naturalistic observation and discussion. Responsive adjustments were made during this phase, such as the addition of structured prompts between Guild topics to generate richer verbal data.

Before the main data collection phase began, an informal conversation with Heli Salomaa provided external validation for the three core workshop principles. Salomaa is a researcher working at the intersection of fashion and game character production, she independently confirmed that flat-to-3D pattern translation, fabric exploration and draping were the most relevant areas of costume design literacy for game character production workflows. While the conversation was informal, this confirmation from an experienced practitioner strengthens the rationale for the workshop content before observation began (H. Salomaa, personal communication, February 2026)

3.2 PARTICIPANTS

Participants and eligibility. Participants were game art students from all years of the bachelor's programme at BUas. Participation was open to anyone with an interest in creating clothing for character design in games, meaning the group was self-selected toward students with pre-existing curiosity about costume or garment construction. Several participants brought prior experience with sewing or fashion, which shaped how they approached the material. Findings therefore speak specifically to how curiosity-driven students engage with costume principles and cannot be assumed to represent the broader game art student population.

Recruitment process. Recruitment reached participants through an open invitation and introduction presentation distributed via Microsoft Teams (Microsoft, 2025), the program's official communication channel, through which a dedicated channel was created to communicate and announce sessions. The invitation framed the Guilds explicitly as voluntary enrichment opportunities with no homework requirements to prevent transforming curiosity-driven learning into academic burden. Students were told the discussion activities were done to generate data for this master's thesis but that it had no grading implications.

Guilds were delivered online-only via Teams as the testrun phase and student agenda analysis revealed it as most accessible for the Friday timeslot. Online-only also removed the hybrid preparation burden. Neither hands-on engagement nor non-verbal behaviour observation was possible during the Guilds, but the setup compensated by acting as the main data collection method that captured student reflections on applying newly acquired costume knowledge to evaluate or reconsider their own character design work.

The workshop was delivered in-person at Buas, as physical material engagement is central to the pedagogical approach and cannot be replicated in an online format. The workshop was offered as complementary to the Guild series but could also be attended as a standalone session. Of the students who expressed interest, six attended in person, representing those with sufficient motivation to engage beyond the online sessions.

Sample characteristics. Participants represented students from across all four years of the game art program, bringing varied levels of technical experience and game development knowledge. While systematic demographic data was not collected, disciplinary background within the program was noted informally during sessions as it seemed potentially relevant to how students approached the material. Whether this proved analytically significant is discussed in section 4.4. Attendance varied across sessions, and the voluntary nature of

participation meant that enrolled students likely represented those with existing interest in character design or costume topics, creating self-selection toward engaged learners.

Sample size justification. The small sample size aligns with design-based research methodology, which prioritizes voluntary enrolment and iterative refinement across small groups over large-sample generalization (Paine & McMahon, 2023; Robinson & Alvarez, 2024). Rather than seeking statistical generalizability, the study aimed for transferability: Merriam (2009) describes transferability as achieved through rich contextual description of the setting, participants and findings, allowing educators in similar contexts to assess applicability to their own situations (p. 227).

Researcher positioning. The researcher occupied multiple roles as workshop facilitator, curriculum designer and observer, Merriam described “observer as participant”, where the participation of the researcher is secondary to information gathering (Merriam, 2009, p. 124). As facilitator and curriculum designer, the researcher designed and delivered content, guided activities and responded to student comments and questions. Simultaneously, the researcher maintained an analytic stance, observing engagement patterns and noting spontaneous connections students made during sessions. The researcher's background in digital fashion design shaped this dual position, as textile and construction expertise informed both curriculum development and data interpretation.

3.3 MATERIALS AND TOOLS

Guild content development. Synthesis of costume design literature (Adams, 2023; Honeycutt, 2021; Jablon-Roberts & Sanders, 2018; Lyons, 2020) and analysis of industry practice (Cork, 2015; Salomaa, 2019) were the foundation of the Guild content development. AI-assisted drafting tools were used to generate initial content structures, which were subsequently verified and refined by the researcher against fashion history sources, drawing on formal training in fashion and textile design and supplementary historical costume sources (Fashion Institute of Technology NYC, n.d.; The Metropolitan Museum of Art, n.d.; Victoria & Albert Museum, n.d.). Four historical periods were selected as most relevant for video game character design contexts.

Guild structure and delivery. The Guilds ran across five sessions: one introductory session establishing shared foundational knowledge, and four historical period sessions each addressing a distinct period relevant to video game character design. Each session alternated between theoretical presentation and in-between discussion prompts, closing with a wrap-up discussion. Content was delivered via PowerPoint presentations and iteratively refined

based on student engagement patterns and topic requests documented during the testrun phase, following Merriam's (2009) principle that qualitative research involves simultaneous data collection and analysis enabling responsive adjustment (p. 165). For full session content documentation see appendix A.

Workshop theoretical introduction. The hands-on workshop opened with a brief theoretical content delivery introducing the participants to fabric jargon (bias, warp, weft) connected to simulation software property sliders (stretch, shear, bending, buckling, weight and internal damping). These properties influence how a fabric behaves digitally, providing a foundation for understanding fold types and drape logic introduced before the activity stations. See appendix B.2.

Workshop activity materials. The hands-on workshop activities used tactile materials selected to create concrete experiential learning opportunities. Kolb's (1984) experiential learning theory emphasizes that knowledge is created through the transformation of experience requiring physical engagement rather than passive observation (p. 21). Similarly, Lyons (2020) documents that kinetic-to-visual learning through material handling creates stronger retention than lecture-based instruction alone. Activities were selected to enable this embodied learning approach, see the appendix B.3 for a comprehensive documentation of activities and materials used.

Personal response sheets. At the end of the hands-on workshop, participants completed a written personal response sheet. The sheet contained several open reflection questions inviting students to articulate what they noticed, what surprised them, and how they might apply what they encountered to their own character design work. Space was provided for additional comments. The written format was selected over a group discussion as this allowed students to pause and write in their own pace without the pressure of sharing it out loud. The response sheets function as qualitative self-report data, capturing student articulation of their own learning rather than researcher observation of it. The full response sheet questions are documented in appendix B.5.

Guild series review. Following the final session of the Guild series, a short follow-up questionnaire of six structured reflection questions was distributed to participants via the Guilds unofficial Discord channel. The questionnaire asked students to reflect on which sessions or topics they found most relevant to their character design practice, what content they would have liked more of, and how the series had influenced their thinking about costume and character design. This supplementary data collection extended the verbal discussion data gathered during sessions, providing post-series reflection that participants may not have

articulated in the moment. Responses are used to support emerging themes in the data analysis and inform the future directions chapter. The full review questions are documented in appendix A.5.

Recording equipment. The online Guilds were recorded through the built-in Teams recording function which captured the PowerPoint presentation, participants and audio. This function also transcribed everything that was said, downloaded separately for data analysis. In combination with the video recording, this enabled post-session analysis of content and student discussions in video and text format. The hands-on workshop was captured through video data using OBS studio (OBS Project Contributors, 2025), an open-source screen recording application. OBS was configured to simultaneously record the facilitator's computer screen (displaying PowerPoint presentation) and the input of a webcam capturing the students seated in the classroom or during hands-on activities, including audio. This dual-capture setup enabled post-session analysis of both the content delivered on screen, the activities and student behavioural responses to that content. Additional audio backup was recorded using a mobile phone with recorder app, positioned to capture student discussion that was not captured by the computer's built-in microphone.

Field note limitations. Merriam (2009) notes that while video and audio recordings can preserve a complete record of sessions, the research should still take highly descriptive field notes to document contextual details and preliminary analytic impressions (pp. 129-130). Field notes were taken during the hands-on workshop session, though the facilitator's active teaching role meant these were necessarily incomplete as interruptions for student questions and demonstrations prevented continuous notetaking. Post-session field notes were therefore written immediately after the session to supplement live observations, capturing preliminary impressions and analytic reflections while recall was still fresh.

3.4 PROCEDURES

Data collection timeline and setting. Data collection of the Guild sessions occurred during five 60-minute Guild sessions conducted between 6 February and 13 March 2026 online in Microsoft Teams. Data collection of the hands-on workshop happened on 4 March 2026 in a classroom at Breda University of Applied Sciences. The classroom provided a presenter screen for the PowerPoint presentations and table seating for 40 students but 3 groups of 9 tables were made to offer enough space for each activity station.

General session structure. Guild sessions followed a consistent format alternating between content delivery and prompted discussion segments. Sessions opened with

a short introduction and agenda overview, moved through alternating presentation and in-between topic discussion prompts, and concluded with a final group discussion and announcement of next week's topics. The hands-on workshop followed a different structure organised around three activity stations: textile exploration, fabric behaviour and draping, and patterning. The workshop opened with brief theoretical framing. Participant groups rotated through stations at 35-minute intervals, with the researcher circulating between groups rather than delivering content from the front.

Student initiated discussion. Responsiveness to student confusion and interest signals was essential to the research aims rather than purely passive observation. This extended beyond structured prompts to student-initiated tangents, which were treated as data rather than interruptions. Franz (2014) notes that the "unfocus" that happens when groups stray from interview protocols, can reveal valuable insights into areas of the topic not previously considered. (p. 1383) Unexpected connections to specific game examples, current coursework or other disciplines were therefore noted as indicators of which costume principles made an immediate impression on students.

Iterative refinement during data collection. Two adjustments were made during the main data collection phase based on emerging observations. In the Guild series, in-between topic discussion prompts were introduced after the second session as online delivery limited naturalistic observation, and the prompts generated verbal data that would not otherwise have been captured. In the hands-on workshop, a challenge extension was added to the patterning activity to accommodate students who completed the foundational task quickly, preventing disengagement among more experienced participants.

Recording procedures. Teams recordings began five minutes past session start to allow for late arrivals and continued uninterrupted through conclusion. The hands-on workshop was captured via OBS studio, recording the facilitator's screen and webcam simultaneously, with a backup audio recorder capturing discussion not picked up by the computer microphone. The facilitator operated all equipment alone. Students were informed verbally at session start that the session was being recorded for research purposes, and no discomfort was expressed across any session. Immediately after the hands-on workshop, approximately 15-20 minutes of post-session field notes were written to capture non-verbal observations that video recording alone cannot fully preserve, moments of apparent confusion or breakthrough (Merriam, 2009, pp. 128-131). Guild sessions were not followed by formal field notes as the Teams transcript provided sufficient documentation of the verbal data accessible in that online format.

3.5 DATA ANALYSIS

Analysis approach and timing. Data analysis occurred iteratively throughout data collection rather than as an afterthought in the post-collection phase. Following Merriam's (2009) principle that qualitative analysis is "simultaneous with data collection" (p. 169), analysis of the previous workshop session informed methodological refinements for the next, showing responsive design wherein emerging patterns guide the following data collection.

Qualitative analysis of observation data. Video recordings were reviewed to assist in recalling patterns in student engagement, moments of comprehension or confusion, document quotes of spontaneous connections to game design contexts, and problem-solving approaches. Analysis focused on three dimensions: behavioural engagement (visible student actions during activities such as material manipulation strategies, collaborative discussion), verbal indicators (questions suggesting comprehension gaps, exclamations signalling breakthrough moments, unprompted connections to game development practice), and peer interaction patterns (moments where students drew each other's attention to specific material properties or observations) and sustained engagement indicators (whether students continued interacting with a topic or material after the session moved on).

Qualitative analysis of verbal data. Thematic analysis following Braun and Clarke's (2006) six-phase framework was used to analyse observation and verbal data across all sessions. Analysis began with purposeful reading of all transcripts and session notes, guided by the research question and concrete reading lenses derived from it, specifically, what engagement with costume principles would look like in student verbal responses and behaviour. Relevant fragments were extracted and transferred to a digital visual workspace (Miro, 2026) where each fragment received an initial code describing what the student was doing with the knowledge rather than simply what topic was being discussed. Codes were then examined analytically by noting patterns, absences and surprises, before being grouped into broader clusters. Clusters were refined iteratively until three to four distinct themes emerged, each capable of carrying a declarative statement as a header. This process followed Braun and Clarke's principle that themes should reflect meaningful patterns in relation to the research question rather than simply frequency of occurrence (Braun & Clarke, 2006).

Guild session transcripts and workshop audio recordings were the primary verbal data sources. Transcription focused on documenting meaningful information as the research questions centred on what knowledge students found relevant (what concepts they reported finding valuable or applicable), persistent confusion (concepts that remained unclear

despite explanations), and envisioned applications (how students described using costume principles in their character design practice).

Triangulation strategy. Findings were triangulated across three data sources: behavioural engagement, verbal data and discussions, following Patton's principle that when multiple qualitative sources converge on the same finding, that convergence itself constitutes evidence of credibility within a purely qualitative paradigm (Patton, 2002, pp. 559-560). Triangulation occurred through cross-referencing: if video showed prolonged engagement with a specific fabric sample, corresponding discussion transcripts were examined for verbal data that manifested in spontaneous comments, connections and reactions, and discussions post-activity were examined for related articulation of principles learned through that tactile exploration. Alternatively, when data sources diverged, this was treated as analytically meaningful and prompted closer examination of what the inconsistency revealed.

Responsive analysis for curriculum refinement. Analysis explicitly served dual purposes: characterizing student learning within completed workshops and identifying content refinements for the following ones. One example of this responsive process was the introduction of in-between topic discussion prompts during the Guild series. After the second session, observation of limited verbal data generation prompted the addition of structured prompts to create deliberate pause points for student articulation. This responsive adjustment reflects design-based research methodology wherein analysis continuously informs design.

Trustworthiness and credibility. Triangulation through multiple data sources (engagement, verbal data and discussions) strengthened confidence in findings when overlapping patterns emerged. Rich, thick description was prioritized: detailed documentation of workshop context, student behaviours during specific activities, and quotes from discussions provides evidence enabling readers to assess interpretation credibility and transferability to their contexts.

Researcher positionality in analysis. The dual positioning as costume design expert and game technology researcher shaped analytic interpretation in specific ways: the researcher could recognize construction thinking that might be invisible to analysts without fashion training while simultaneously recognizing vocabulary translation moments that might be missed by analysts from pure fashion backgrounds.

This positioning carries both strengths and limitations. Strengths include ability to distinguish meaningful engagement from superficial participation and to recognize subtle conceptual breakthroughs. Limitations include potential confirmation bias: as curriculum

designer and facilitator, the researcher may favour evidence supporting pedagogical effectiveness while overlooking confusion or disengagement. To mitigate this risk, analytic attention was directed toward divergences such as moments when data sources conflicted, when students expressed confusion, or when activities did not produce expected outcomes.

Additionally, as the curriculum designer responsible for the sessions being observed, analytic interpretation carries the risk of investment in the content itself influencing what the researcher registers as meaningful engagement. This was mitigated by prioritising student-generated language over researcher inference, and by documenting moments of disengagement or confusion alongside moments of resonance.

4 FINDINGS AND DISCUSSION

Analysis of data collected across five Guild sessions and a hands-on workshop produced four recurring themes. Each theme reflects a distinct pattern in how students engaged with and translated costume design principles through the lens of their own discipline and discipline role.

The four themes that emerged from analysis share a common thread. Students did not receive costume knowledge passively but translated it through their existing discipline role, whether as a character artist, concept artist or rigger. That translation is where the engagement happened. It is what generated new ideas, shifted how students evaluated their own work and connected unfamiliar principles to problems they already recognised. The themes that follow each capture a different part of this process. Supporting quotes for each theme are documented in Appendix C.

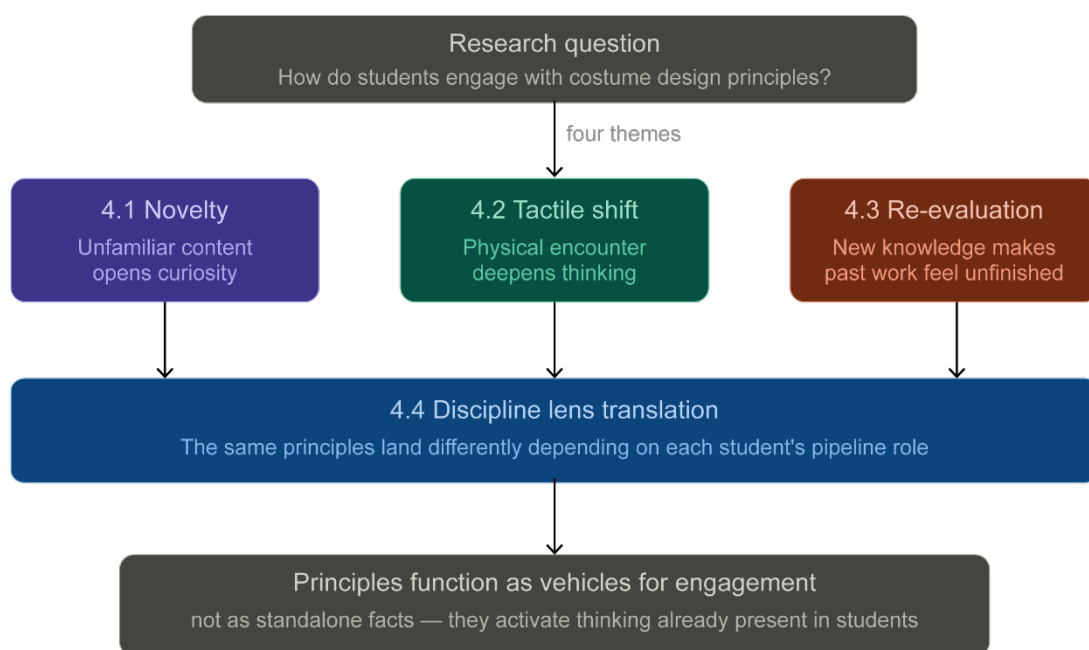


Figure 5 - Four themes emerging from data analysis
Showing how costume principles activate different types of student engagement.

4.1 SOMETHING I'D NEVER SEEN BEFORE MADE ME WONDER WHAT ELSE I DON'T KNOW

It's not really a material I thought of to be used in this way — but from a worldbuilding standpoint, if a fantastical creature could be hunted, perhaps materials like that can show up in fashion in similar ways." — P2

An emerging pattern in the data shows that unexpected or unfamiliar content consistently generates stronger engagement than familiar content. Where students could recognise what they were encountering, responses tended to stay at the level of confirmation. But when content surprised them, like an unfamiliar material, unexpected historical fact or a principle they hadn't considered, students moved beyond recognition into wider curiosity, application and idea generation.

The strongest example of this is the mushroom leather that students encountered during the hands-on workshop. Multiple students named this as the most memorable moment across all sessions. One student described it as something they had never encountered before, reflecting that it made them curious about what other textiles they might not know yet. Another described that learning the material exists, "*blew their mind*". A third started brainstorming application ideas like using it to texture fantasy forest characters. And another student translated the encounter immediately into a design application, brainstorming how the material could be used to texture fantasy forest characters. What is notable is that these four responses were different in kind: one moved toward curiosity, one toward lasting impression, and two towards immediate application and translation. This suggests that a single unfamiliar material can activate multiple types of engagement simultaneously.

The data shows a split in how students applied new knowledge to their worldbuilding, some students extended existing projects with new knowledge, while others generated entirely new ideas. For students who extended their work, the Guild or workshop content gave them something specific to add or change to a character or world they already had. One student applied the new knowledge about how silk is produced to their moth lady character, realising the material could communicate that other moths had been killed to supply her clothing, turning a fabric choice into a narrative one. Another connected the content about horse barding to a hippogriff character they had already designed and began considering what barding on the wings might look like. The same student also identified an application for a larger biopunk world they were developing, finding that the concept of living clothing that connects to the wearer fit naturally into their setting.

The second application visible in the data is when new knowledge made students generate entirely new ideas. The strongest example of this came from a student who had just learned corset boning was historically made from whale baleen. Finding this unexpected, they immediately extended the principle into speculative territory, proposing that in a fantasy world, a hunted creature could supply similar materials for fashion in the same way. Another student was inspired by programmable material or shape and colour changing fabrics

and began imagining it could change with a single thought or emotion. A third student arrived at a different kind of generation as they reflected on how the sheer volume of fabric in a toga reaches a point of deliberate impracticality, which they thought felt ironic and even purposefully satirical, which can be used as a worldbuilding rule to make fun of the higher class.

Novelty can be seen not just as a pleasant surprise but as a pedagogical entry point, as something that opens a door rather than simply delivering a fact. What the data suggests is that unfamiliar content activated a different kind of thinking in students than familiar content did. Rather than confirming what they already knew, novelty seemed to create a gap between what students thought they understood and what they were now encountering, and it was in that gap that curiosity, idea generation and connection-making happened. It seems that for these students, not knowing something was not an obstacle to engagement but a condition for it.

This aligns with Adams (2023) and Lyons (2020), who both argue that hands-on material engagement creates stronger retention than passive instruction. However, what the data here adds is a nuance neither source directly addresses: it was not simply the handling of materials that generated engagement, but the encounter with materials that were genuinely unfamiliar. Novelty seems to be doing work that familiarity cannot.

However, not everything needed to be novel to be valuable. New additions to familiar content, and the reasoning behind why things looked or worked the way they did, also offered students important foundations, giving them vocabulary and frameworks to develop their ideas further, as described in the sections that follow.

4.2 FROM TOUCHING FABRIC TO RETHINKING DESIGN

"It got me to consider the weave. I never cared about the weave before. Some of the weaves could not be found in the same material, so we had to check on other materials. And that was fun."— P2

Physically handling materials seemed to do something that information alone could not. Where the Guild sessions introduced students to costume principles through slides and discussion, the hands-on workshop prompted a different kind of response, where students began asking questions, making unexpected connections, and translating what they were touching into their own design contexts.

This was visible in how students responded to the fabric exploration station. One student noted that physically handling different materials prompted them to start considering

thread construction, which was something they described as never having thought about before. The activity had been designed so that certain weave types could only be found across different fabric samples, requiring students to search and compare rather than simply observe. It seems this active searching and comparing was what made the weave stick. A different kind of shift appeared during the draping station, where one student made plans to continue the practice independently, wanting to get a smaller mannequin to try draping at home as it's easier than a full-size one.

Perhaps the clearest example of unprompted creative and technical translation happened during the draping activity, one student spontaneously asked themselves "*what fabric would a magician use?*" and began working through the fabric property sliders to spec out what a magician's reveal cloth would need to be: low bending resistance to show contours, a slippery surface for a smooth reveal, without being prompted to do so. The physical act of draping seemed to trigger this as the encounter with fabric in motion activated a design problem the student then immediately tried to solve.

During the hands-on activities, students also made references to game design pipeline contexts and connections emerged to other concepts outside of this context as well. A student was amazed by the coloured yarn bar on the self-edge side of the Gobelin jacquard fabric, which made them think of RGB and connected it to CRT televisions. While another noticed the undyed calico fabric looked just like the default Substance Painter cotton texture. A third student wondered how all the different fabric samples would feel when sewed together and worn as a shirt and later connected the way a fabric draped to the similarity of chainmail draping. In each case the connection was unprompted and immediate and the physical encounter with material activated a reference from the student's existing visual vocabulary as a game artist.

Physical activities and tactile experiences seem to function as a catalyst rather than simply a delivery method. What the data suggests is that handling materials activated a mode of thinking that slides and discussion did not, characterised by spontaneous questioning, unexpected connection-making and immediate creative translation. It seems that when the student uses their senses while encountering something unfamiliar creates a different kind of attention than the mind receiving information, and it was in that heightened attention that some of the richest engagement in the dataset occurred.

This supports what Adams (2023) and Paine and McMahon (2023) describe as the value of hands-on engagement over lecture-based knowledge transfer, and aligns with Kolb's (1984) experiential learning cycle, which positions concrete physical experience as the

entry point into deeper conceptual understanding. What the data here extends is that physical experience does not only precede conceptual shift, it sometimes triggers it unexpectedly, through encounters students had not anticipated and connections they did not set out to make.

Several students noted that physically exploring textiles, patterns and draping were a really fun and playful way of getting more familiar with these topics, helpful for their future work and gave them new understanding. This appetite for hands-on engagement led multiple students to independently request more applied and practice-based sessions, expressing a desire to start building the translation bridge between new knowledge and their own work. That students arrived at the end of a single introductory session already asking for more suggests that hands-on engagement does not just support learning, it generates demand for it, which has direct implications for how costume design literacy is positioned within game art curricula.

4.3 NEW KNOWLEDGE MADE PREVIOUS WORK FEEL UNFINISHED

"I've made a design for that era before — they were relatively colourfully dressed. But having learned about how expensive specific colours of dyes were, I think I might want to revisit that."
— P2

Before encountering fashion history in the Guild sessions, several students described their approach to designing costumes as “vibe-based” and taking costume choices at face value, for example when choosing a material and defaulting to “hard stiff” or “default loose” or choose lots of colour because “it looked fancy”. What the data suggest is that engagement with costume principles introduced a new layer of thinking that starts to give students structural frameworks. Students began to understand why it looked that way, and what it would communicate to the person viewing it.

This shift appeared most clearly in how students consider which fabric to use. A student reflected on how fabrics can be used to have its own purpose instead of functioning as purely a visual texture. The material a garment was made out of, could be designed for specific purposes, like garment-specific or protective against the weather. This reframing transformed fabrics as just an aesthetical choice, into deliberate decisions for narrative and storytelling. It seems that when students start to understand *why* something looks that way or is chosen, rather than simply taking it at face value, changes how students make and evaluate their own design choices.

For some, this manifested as retroactive re-evaluation. One student described choosing very colourful fabrics for a renaissance-fantasy projects zombie servants, without considering if these vibrant dyes would've been available in these large quantities. Knowing how expensive specific colours of dyes could be in renaissance periods, especially purple, made them reflect on their choice that the colours could have been chosen more aligned with fashion history. The knowledge did not simply add historical accuracy, but it introduced a new standard against which previous work felt incomplete. This pattern, where new frameworks caused students to look back at finished work with a more critical eye, was present across several moments in the data and suggests students were not passively receiving new information but actively using it to evaluate their own work.

Perhaps the most interesting response came from a student who came into the Guild session confused about creature adapted clothing, as often a hole is just cut to make space for wings or tails, but that didn't make sense to them, and left the session with new considerations to start thinking deliberately about what needs to be changed or adapted for a creature to actually be able to wear it. This could suggest that costume knowledge doesn't always introduce new things but sometimes it gives students vocabulary and framework to fix something they couldn't before.

Honeycutt (2021) argues that once viewers understand clothing as a coded system, they begin reading garments for meaning automatically. What the data suggests is that this shift can happen in reverse too: once students understood the codes, they began reading their own previous design decisions against that standard and finding them unfinished. This connects to Jablon-Roberts and Sanders (2018), who describe how costume communicates social status through shared cultural codes, as the students who reconsidered dye colours or creature adaptations were essentially recognising that their earlier choices had not been making deliberate use of those codes.

It is worth noting that some students did not have projects they can revisit, as they were earlier in their education and have not made many costume-focused projects yet. For these students, the shift was more forward-looking as they expressed clear intentions to apply this knowledge in future work, considering how fabrics, dyes, materials and construction can be used to tell a character's story more deliberately, for example through weathering, informed material choices, or thinking carefully about what a character or creature's physical form actually requires.

4.4 THE SAME PRINCIPLES, DIFFERENT TRANSLATIONS: DISCIPLINE ROLE AS READING LENS

"Panels, long coats, accessories — all of these are harder to rig because of layering, overlap and different ways someone might have to move. A rich person in a sci-fi universe or a soldier would be much easier to rig than a poor person dressed in hanging clothes." — P7

Before going into what each student did with costume knowledge, it is worth mentioning that they did not make the same translations. The data suggests that the same content, historical periods, fabric behaviour, construction logic, all landed differently depending on who received it and what role they were training for in the game development pipeline. Students engaged most deeply with principles that map directly onto their existing discipline role.

This was the most visible in how the rigger of the participant group engaged with materials and fabric behaviour. They reflected on new considerations about how material should bend and shape depending on the fabric for realistic deformations and began thinking about what a special technical setup for smart accessories might require.

A similar pattern appeared with concept artists, but the translation moved in a different direction. Where the rigger thought in terms of physical behaviour and technical setup, concept artists tended to translate costume knowledge into worldbuilding and visual communication. They mentioned construction knowledge as valuable, that in a time where people had to make their own clothing, stitching can showcase that. Following the same principle, in a world where everyone has radically different body shapes, it's important to know how clothes are made. Or how there is so much subconscious information in fabric dye choices, that even when you don't know that purple is an expensive colour, you still expect rich people when you see them in purple.

Character artists showed a third type of translation, closer to physical form and construction logic. For this group of students, costume knowledge connected directly to how garments are built and simulated. Understanding how flat paper patterns connect to make a three-dimensional shape, or how fabric behaviour in simulation software affects where folds land, gave them a more reliable foundation before sculpting. One student described looking up real historical patterns for a garment they wanted to make a historical, believable garment that looks like it was worn back then and finding that working from accurate construction logic reduced the amount of corrective sculpting needed afterward.

What these patterns suggest is that costume principles do not function as standalone facts that students either take with them, or don't, but rather function as lenses. The lens that each student looks through is shaped by their professional identity and pipeline position, such as which problems their discipline asks them to solve. Each principle is embedded in every moment of translation, surprise, worldbuilding or priority shift, they all happened through a principle. It seems that principles are the vehicle for engagement, not the subject of it.

Makryniotis (2018) documents communication barriers between fashion professionals and game artists as stemming from different vocabularies and ways of understanding the same information. What the data here suggests is that a similar dynamic exists within game art itself: different discipline roles produce different translations of the same costume knowledge, shaped by the problems each discipline is trained to solve. This extends Makryniotis's finding beyond the fashion-game boundary and into the game art pipeline itself, where the same content lands differently depending on who is receiving it.

It is worth noting that not all principles resonated equally across disciplines. Semiotics, meaning the reading of costume as social and narrative communication, was the principle least explicitly named by students in relation to their own work. This may reflect the pipeline position of the participants, who were primarily character and concept artists rather than narrative designers or art directors. However, what is interesting is that semiotic thinking was reached organically through other subject areas such as historical and material content. Students independently arrived at conclusions about how colour communicates wealth, how impractical clothing signals status, and how wearing more layers signal you are in a higher position than the opposite. This suggests that while students may not yet see semiotic thinking as part of their role, their capacity for it is present and can be activated through historical and material content.

4.5 HOW STUDENTS EXPERIENCED THE LEARNING FORMATS

Beyond explaining what students learned, it is worth examining how they experience the formats themselves. After the Guild and workshop, a request to give feedback on the structure, topics and format was given. Understanding how students experienced the format is relevant because perceived experience shapes whether students engage with content at all, and because student responses to the format itself provide data for future iterations.

Students responded well to the way the Guild sessions were structured, particularly the alternation between content and discussion prompts, the historical progression

across sessions, and the variety of topics. One student described the historical framing as their preferred way of receiving information, valuing the reasoning behind how things developed rather than just the facts themselves. Another student reflected that going through the history of clothing helped them understand how things came to be the way they are now. And someone else said that the variety of topics each session kept them interested and created a sense of not wanting to miss out on new information. The in-between prompts were specifically named as useful for keeping students actively engaged rather than passively listening. A student mentioned this helped them stay active listeners, and another mentioned the structure of alternation between information and prompted questions was enjoyable.

Student responses to the workshop format pointed to the physical and social setup as key to what made it work, a student mentioned it was fun and chill which helped them collaborate and made experimentation feel less overwhelming. This student named the social atmosphere as what made them comfortable enough for open experimentation. Another student mentioned the setup was a fun and playful way to get more familiar with the topics, and someone else mentioned this exploration of topics was interesting and helpful. The variety of materials, activities and the tactile nature were highlighted as particularly effective and students named specific activities as useful for future work, describing how physical material exploration helped them translate fabric knowledge into digital practice and prompted considerations they had not encountered before.

The positive responses about the format design suggests that the complementary design of Guilds and workshop seems to have created different but reinforcing conditions for engagement. The Guilds created theoretical grounding and appetite, and the workshop gave that appetite somewhere physical to go. Together they seem to have produced the engagement described in the previous four themes. This suggests that the format was not incidental to the findings but generative of them and the conditions the sessions created shaped what students were able to notice, connect and articulate.

There were also some things that didn't work such as the group assignment booklet which was given to fill in during the activity which was meant to function as a prompter for students to get a discussion going. A student mentioned this was difficult to complete while also focussing on the experimentation of the activity and mentioned they struggled to find a moment to write their thoughts in it. The second thing that a student mentioned that they didn't like was the forced groupwork for the fabric exploration activity, while another student mentioned they liked the collaboration. This tension suggests that the social setup of activities affects engagement differently across students, and that offering individual as well as group

options may better accommodate different working styles. These friction points suggest that some of the data collection tools need to be integrated differently into the activities for future iterations.

Multiple students independently requested more hands-on and applied sessions for the following half-semester, expressing a desire to move from receiving knowledge to actively using it. Their suggestions consistently pointed in the same direction: applying costume principles to actual student projects rather than exploring them in isolation. Ideas included breaking down existing game character costumes, creating a garment together during a session, walk-in style work discussions, and exploring how specific environmental conditions like extreme cold or underwater settings affect clothing and weathering. Whether the format was a workshop, a group brainstorm or a weekly challenge, the common thread was a need for a translation bridge between what students had learned and what they were making. The format was not incidental to the findings but generative of them and the conditions the sessions created shaped what students were able to notice, connect and articulate, and that has direct implications for how future iterations are designed.

4.6 SYNTHESIS

Taken together, the four themes suggest that costume design principles are most useful not when treated as content to deliver, but as material to translate. Students engaged most deeply when principles created friction with what they already knew or assumed, and it was in resolving that friction through their own discipline lens that shifts in thinking and decision making occurred.

This pattern held consistently across all three data sources. Behavioural observations during the hands-on workshop showed students physically working through that friction: searching across fabric samples to find a weave, adjusting drape until the fold behaviour made sense, assembling pattern pieces until the flat-to-3D logic clicked. Verbal data from Guild sessions and workshop discussion captured the moment of translation as it happened, one student in a several sessions connected baleen to fantasy creature materials, then reconsidered their historically inaccurate dye choices, and during the hands-on workshop spontaneously specced out a magician's reveal cloth using simulation slider logic. Personal response sheets confirmed that these moments carried beyond the session itself, with students naming specific materials, activities and principles they intended to bring into their future character design work. When all three sources pointed to the same principle or moment, that convergence is what the themes in this chapter are built from.

This answers the research question directly. The costume design principles that proved relevant for game character design are construction logic including patterning and flat-to-3D translation, historical and material context, fabric behaviour, and semiotic thinking. Students engaged with and applied these principles by translating them through their existing discipline role. What made the formats work was that they completed each other: the Guild sessions built theoretical grounding across historical and material content, and the hands-on workshop provided the tactile physical engagement that online delivery could not, together covering what neither format could have achieved alone.

5 CONCLUSION

This study asked: what costume design principles are relevant for game character design, and how do students engage with and apply these principles in workshop contexts?

The answer is that students engage with costume principles by translating them through the lens of their own discipline and discipline role, and that this translation is where learning happens. It generates new ideas, shifts how students evaluate their own work, and results in more deliberate creative decisions. The principles that proved most relevant across all participants were construction logic, historical and material context, fabric behaviour, and semiotic thinking, though semiotic understanding emerged organically rather than through direct instruction.

Before this research, the documented problem was friction at the boundary between fashion professionals and game artists, stemming from different vocabularies and ways of reasoning about the same information (Makryniotis, 2018; Salomaa, 2019). What this study found is that the same dynamic runs through game art education itself. The same costume content landed differently depending on discipline role: the rigger translated material behaviour into deformation logic, concept artists translated it into world- building and visual communication, and character artists connected it to construction and simulation workflows. The gap is not only between disciplines but within the pipeline, and that has direct implications for how costume design literacy should be taught.

This study also adds a distinction not addressed in prior work on hands-on learning. Adams (2023) and Lyons (2020) both establish that physical material engagement produces stronger retention than passive instruction. What the data here suggests is that novelty does specific work that familiarity cannot. Students who encountered genuinely unfamiliar materials showed the broadest and most sustained engagement, moving simultaneously toward curiosity, lasting impression, and immediate design application. Familiar content still mattered as it gave students frameworks and vocabulary to develop their ideas further, but unfamiliar content created the friction that activated thinking.

A final observation concerns semiotics. Semiotic reasoning was not taught as a standalone topic, yet students arrived at semiotic conclusions independently through historical and material content. They identified how colour communicates wealth, how impractical clothing signals status, and how layering encodes hierarchy, without being directed toward those conclusions. This suggests that semiotics functions most effectively as something woven through other content rather than introduced as its own topic.

Together, these findings provide the foundational understanding the field was missing. No prior research had systematically identified which costume design principles resonate with game art students or documented how those students engage with interdisciplinary textile knowledge in workshop contexts. This study does not complete the pedagogical bridge but establishes the conditions under which that bridge can be built: principles that create productive friction, formats that meet students where their discipline already sits, and room for translation rather than transmission. The constraints of this study and the directions it opens are addressed in the following chapter.

6 LIMITATIONS AND FUTURE DIRECTIONS

This chapter reflects on the constraints of the current study and identifies directions for future development. Section 6.1 addresses methodological limitations that shape what can be claimed from the data. Sections 6.2 and 6.3 outline recommendations for the next iteration of the workshop and a broader research agenda for the field.

6.1 LIMITATIONS

The findings of this study carry several constraints that shape what can be claimed from them.

The participant group was self-selected, meaning the data speaks to motivated, curiosity-driven students rather than the broader game art student population.

Engagement patterns documented here represent what is possible under favourable conditions, not what is typical. Year-level variation also affected data richness, as lower-year students gave flatter responses and articulated connections less readily than higher-year students, suggesting that design maturity plays a role in how productively students engage with interdisciplinary content.

The researcher occupied simultaneous roles as facilitator, curriculum designer and observer, which carries confirmation bias risk. This was mitigated through triangulation across behavioural observations, verbal data from sessions, and personal response sheets, and by tracking moments of confusion and disengagement alongside moments of resonance. The risk cannot be fully eliminated in single-researcher design.

The online format of the Guild sessions made naturalistic observation impossible across five of the six data collection sessions. Signs of confusion or disengagement were only captured when students named them verbally. Single-researcher data collection in the hands-on workshop also produced moments of incomplete data, the most significant being the magician moment, where a rich spontaneous design application could not be fully transcribed because multiple students spoke simultaneously. The group assignment booklet contributed to this, as students struggled to write while also engaging with the activities, meaning some responses that would have been data were never recorded.

6.2 RECOMMENDATIONS FOR FUTURE ITERATIONS

A second observer would allow the researcher to focus on facilitation while someone else captures non-verbal engagement, station-level dynamics, and the kind of simultaneous speech moments that produced incomplete data in this study. The group assignment booklet should be replaced with shorter card-based exercises that sit within the activity rather than alongside it. For the mystery fabric station specifically, answer cards would reframe the task as guided deduction and reduce the cognitive split between doing and documenting. The audio backup setup also needs improvement: a single mobile phone positioned to catch group discussion was not sufficient when multiple students spoke at once, and two dedicated recording positions per station would reduce data loss.

The workshop timing of 35-minute station intervals worked well for the introductory format this workshop was designed to be. For follow-up sessions that move beyond introduction into deeper experimentation, more time per station would allow students to explore further, and splitting across two sessions is one practical way to achieve that without compressing the activities.

Future content should build novelty in deliberately. The data showed that genuinely unfamiliar materials generated broader engagement than familiar ones, which means testing assumptions about what students already know rather than assuming what will feel new to them. Future curriculum iterations should also include pipeline-role entry points, offering riggers, concept artists and character artists different applications of the same shared content, since the same principles land differently depending on who receives them.

The most pressing structural recommendation is curriculum integration. Voluntary enrichment activities consistently lost attendance to coursework pressure even among students who expressed genuine interest. The content needs a place inside the assessed curriculum to reach students reliably and allow observation of how principles develop with repeated exposure.

6.3 BROADER RESEARCH AGENDA

The immediate next step is a follow-up study tracking whether the engagement observed in sessions translated into different design decisions in students' subsequent project work. This study cannot answer that question.

Beyond that, the same workshop format should be tested at other institutions to assess whether the discipline-specific translation patterns documented here are specific to

BUAs, or appear more broadly across game art education. Year level also deserves its own dedicated investigation rather than treatment as a contextual footnote, as understanding when in a game art education costume design literacy is most productively introduced has direct implications for curriculum placement.

This research and Salomaa's ongoing practitioner work are converging on the same problem from different directions. The field is ready for systematic integration of costume design literacy into game art education. What it still needs are the structured pedagogical methods to do it. That is where the next study begins.

APPENDIX A: GUILD SERIES DOCUMENTATION

Full presentation materials and session recordings are available upon request from the researcher at vdwel.imara@outlook.com.

A.1 SESSION OVERVIEW AND STRUCTURE

The costume design Guild ran across five online 1-hour long sessions, delivered through Microsoft Teams. During the Guild session, the content was delivered through theoretical presentations using costume principles applied to historical periods to explain how each principle manifested throughout time. Each session opened with a short intro and the agenda, alternating between presentation and in-between discussion prompts that close or open the next subject. The Guild session closes with a wrap-up discussion and a preview of the following session's topic.

A.2 INTRODUCTORY SESSION CONTENT

The introductory session-oriented participants to the series by establishing shared foundational knowledge. It opened with an explanation why costume design literacy matters for game artists across all visual styles, not only in high-fidelity or realistic game contexts. From there, foundational concepts were introduced: visual language, material behaviour, construction logic, details and layering, to give participants a shared vocabulary for the sessions ahead. The session also covered creature and beings-stereotypes across time periods, period mixing in existing games, and a brief overview of tools used for character costume creation. The session closed with the wrap-up discussion and a preview of the following session.

A.3 HISTORICAL PERIOD FRAMEWORK

The remaining four sessions each addressed a distinct historical period using a consistent content framework. Four periods were chosen as most relevant to video games: Medieval & renaissance, Victorian & Industrial, Ancient to folk & cultures, and Modern to cyberpunk/sci-fi. In each session, the same framework explained the costume principles through the lens of the period and was as follows: each session started with a period explanation and game examples of games inspired by that period. Then silhouette & key garments, followed by social hierarchies and how that could be recognised through clothing. Then materials, textiles and dyes, what was available shaped what was worn. Followed by construction & details. And each session ended with a special topic relevant to the period. Medieval & renaissance's special topic was horse tack and mounts (rideable creatures), Victorian & Industrial had a corsetry deep dive, Ancient to folk & cultures special topic was creature adaptations, how do you design adaptive clothing when there's more arms, a tail etc., and Modern to cyberpunk/sci-fi went into speculative fabrics, how

do you design a material that has no grounding in real life yet. And all sessions ended with the wrap up and discussion.

A.4 ITERATIVE REFINEMENTS

After the second session, in-between topic discussion prompts were introduced. Online delivery made naturalistic observation of non-verbal engagement impossible, and the wrap-up discussion alone generated insufficient data. Structured prompts created deliberate pause points that produced richer verbal data throughout each session.

A.5 GUILD SERIES REVIEW

A request to review the Guild series was shared with participants at the close of the final session. Questions:

1. Which Guild was most useful for you personally?
2. Have you used anything in your work yet?
3. What topic did you find most surprising/unexpected?
4. (hypothetically) If I run this again next year, what would you change?
5. What worked really well and what didn't?
6. What would you like to see in the following half-semester (topics, activities etc)?

APPENDIX B: HANDS-ON WORKSHOP DOCUMENTATION

B.1 WORKSHOP OVERVIEW AND STRUCTURE

The workshop was designed around three activity stations, each focused on a different costume principle. Three groups of six students would rotate through the stations simultaneously in 35-minute intervals with the researcher circulating between the tables to answer questions. On the day of the workshop, nineteen students had signed up but six attended. Rather than splitting into rotating groups, participants moved through the stations chronologically as a single group.

B.2 THEORETICAL INTRODUCTION CONTENT

The workshop opened with a theoretical introduction to the key terms of fabric: warp, weft and bias, which were foundational knowledge to understand the next topic of fabric physical property sliders as used in pattern-based simulation software such as Marvelous Designer and CLO3D. The sliders stretch, shear, bending, buckling, weight and internal damping were explained using real life fabrics (a knit and raw denim). The introduction closed with a run through of fold types (pipe-, drop-, diaper-, spiral-, zigzag-, and half-lock folds) and what influences their drape, followed by an overview of different pattern making approaches. The activity stations were then explained before students gathered at their first station.

B.3 STATION DESCRIPTIONS

Fabric exploration station. The fabric exploration station was set up with thirty fabric samples, each labelled with fibre content, construction type and coloration method, alongside five unlabelled white fabric samples identified only by burn test results, a magnifying glass, and the Clothing Technology reference book (Eberle et al., 2014). Participants were invited to explore the fabrics on the table, using the burn test results, the reference book, and comparison with the labelled samples to identify the five mystery fabrics. A challenge exercise asked students to determine the simulation slider values of a chosen fabric based on its observed behaviour, stretch and drape. Participants did not reach the challenge exercise within the time given.

Draping and fold challenge station. The draping station was set up with two mannequins, three mounted spheres, pins, and a variety of pre-cut fabrics in draping-appropriate sizes. Participants draped fabric across the mannequins and mounted spheres, experimenting with fold behaviour guided by challenge cards, each illustrating a different fold type. In the group assignment booklet, participants were asked to drape the same form in two different fabrics and record what influenced the difference, using simulation slider terminology where possible.

Pattern understanding station. The pattern understanding station was set up with pre-cut pattern pieces of basic block garments, tape, scissors, and additional paper and pencils for the challenge extension. Participants assembled the pattern pieces into three-dimensional garments using tape, experiencing the flat-to-3D translation that underlies garment construction. A challenge extension invited students to slash and spread pattern pieces, experiencing how patterns can be edited to change garment volume and shape.

B.4 WORKSHOP END

At the workshop's close, participants gathered for a short group discussion and sources for further exploration were shared. Participants were then asked to complete the remaining sections of the personal response sheet before handing it in.

B.5 PERSONAL RESPONSE SHEET QUESTIONS

The personal response sheet opened with participant information: name, year of study and discipline. The pre-workshop section contained three self-reported confidence questions: whether participants could identify the textile composition of their own clothing (no, partially, yes), how familiar they were with fabric types and their properties and behaviour (unfamiliar, somewhat familiar, very familiar), and whether they had previously considered how fabric behaviour affects character design (never, sometimes, always). These same three questions were repeated post-workshop to capture any shift in self-reported confidence. The post-workshop reflection section asked two open questions: what one thing they would bring into their game work from the session, and anything else they wanted to share about how they experienced the workshop. The reverse side of the sheet was left open for free responses, including thoughts, sketches, questions, surprises, confusions, and areas participants wanted to explore further.

APPENDIX C: SUPPORTING QUOTES BY THEME

C.1 Findings & discussion 4.1

P8 — Hands-on workshop personal response sheet

"What surprised me was the mushroom leather to be honest, I've never seen anything like it before, or even if I did I never knew that it was made from mushroom. This makes me more curious about all other types of textiles that I probably never heard of before."

P1 — Hands-on workshop, during the textile exploration activity

"Mushroom fabric exists, good for texturing fantasy forest characters"

P2 — Hands-on workshop personal response sheet

"Learning about whacky stuff like mushroom leather was particularly cool, and it is that kind of detail that sticks with me."

P1 — Guild feedback request

"That mushroom fabrics exists, blew my mind"

P3 — Guild 2, Medieval and Renaissance

"I was wondering if maybe I could make it from silk, because after all, she's a moth lady. Maybe she can use that because she attracts other moths. And that insight to like, I was thinking now I can apply it to like she kills other moths to make her own clothing or stuff like that. I really like that part and I would love to see other stuff like that."

P2 — Guild 2, Medieval and Renaissance

"I also recently made a character with a fantastical animal companion, and I drew like a courier that rides a hippogriff, so like a like a postman on a hippogriff. And I made a saddle for that. It did give me the fun idea to maybe, not for this courier specifically, but for if I want to make like a more of like a knight on a hippogriff back, that it would be fun to maybe look into at least like the barding, like the fancy cloth that's on it. That could be a fun little exercise, as maybe there's something on the wings as well that could be interesting."

P2 — Guild 5, modern to future and sci-fi

"Well, I mean, because of the setting I'm working on, the idea of living clothing is something I find very fun and think could be very fitting for my more bio-punk setting. So that is definitely something I want to look into. or like pay a little more closer attention to, especially the idea that it may be almost like connects to the person in and of itself. As a concept artist, by the way, for the record."

P2 — Guild 3, Victorian and industrialisation

"I found it really interesting about the use of baleen. It's not really a material I thought of to be used in this way. And I just find it interesting from a world building standpoint that perhaps if a more fantastical creature would be hunted that perhaps materials like that can show up in fashion in this way as well."

P4 — Guild 5, modern to future and sci-fi

"I would like to use some kind of material on the clothing which would portray the emotion or some kind of connexion with the player. So basically, it would like change colour or shape based on how the character feels at the moment and that could nicely portray to the player the emotions of like their character or maybe the enemies or something."

P4 — Guild 5, modern to future and sci-fi

"Oh, it's programmable material, I really like the idea of like being able to manipulate your clothing by, for example, having a chip in your head and like just thinking about changing the shape in your right hand or something like that. So like... The mechanical aspect of cyberpunk combined with the programmable materials could be really interesting."

P5 — Guild 4, ancient to folk and cultures

"It feels like very ironic and even purposefully satirical. It feels like someone would design it, would make that as a rule specifically just to make fun of the higher class. I find that very interesting that it's like actually a costume"

C.2 Findings & discussion 4.2

P7 — Hands-on workshop, after the textile exploration activity

"It got me to consider the weave. I never cared about the weave before. Some of the weaves could not be found in the same material, so we had to check on other materials. And that was fun."

P9 — Hands-on workshop personal response sheet, about draping

"I really liked this, I might get a smaller mannequin to try this at home soon, easier than real size."

P2 — Hands-on workshop, during the draping activity

"which fabric would a magician use? (then went on to describe the fabric physical property sliders and what it needs to be for smooth reveal, interesting contouring of object etc, data incomplete as several students talked over them)"

P11 — Testrun Guild, textiles, materials and dyes

"OMG it's RGB, if you look at this you can see red yellow..., I looked closer and I was like wait a second! It looks like an old CRT television."

P12 — Testrun Guild, textiles, materials and dyes

"This looks like the default Substance painter cotton texture."

P1 — Hands-on workshop, during the draping activity

"It really looks like chain mail with that fabric."

P1 — Hands-on workshop, fabric exploration activity, about the fabric swatches

"I wonder how all the fabrics would feel if you would wear it as a shirt."

P9 — Hands-on workshop personal response sheet

“Really fun/playful way to get more familiar with these topics.”

P1 — Hands-on workshop personal response sheet

“1st time working with fabrics (draping) was fun and also challenging and experiencing working with various fabrics was a learning-full experience”

P2 — Hands-on workshop personal response sheet

“I like getting to explore this more specific topic this way. It is super interesting and I find it super helpful. Particularly in designing and portraying outfits.”

P2 — Hands-on workshop personal response sheet

“It was a lot of fun. The different materials were very cool and interesting. The folds and drapes were helpful for my future drawings and the patterns were fun.”

C.3 Findings & discussion 4.3

P2 — Guild 2, Medieval and Renaissance

“I had a personal project going that is like Renaissance fantasy, where basically there's like zombie servants, but they would be dressed very fancily. And I've made a design for that before — they were relatively colourfully dressed. But having learned about how expensive specific colours of dyes were, I think I might want to revisit that in general.”

P2 — Hands-on workshop personal response sheet

“Learning more about clothing in general makes me feel more confident. Where previously it always felt more vibes based, now I get to understand this a bit better”

P7 — Guild feedback request

“I always had it in my mind that different materials will affect how clothing should move on a character as well, but I feel like very often I did not have the correct image in my head so I ended up going to a default hard stiff or a default loose and no in between. It was nice to get some nuance for physics.”

P6 — Guild 5, modern to future and sci-fi

“I also like the sci-fi fabrics. I never really thought about it that way. So yeah, I think that's really cool. You can, yeah, think how fabric can have its own purpose instead of just serving as fabric.”

P8 — Hands-on workshop personal response sheet

“I will for sure explore more fabric types in the future and actually think about them while designing. What type of fabric is good for what weather and what type of garment.”

P8 — Guild feedback request

“When I will start making a new project with clothing I will definitely be coming back to the presentations to remind myself of what I should be looking for”

P4 — Guild 4, ancient to folk and cultures

"I was really surprised by the creature adapted clothing because I have never really thought too much in depth how the clothing would need to be adapted or changed for the creature to actually be able to wear it. And honestly, I really thought that you can just like... cut the hole in there for the wings, for example, but when I was thinking about it in the morning, it just didn't make sense, and now I got the answer, so that was really nice."

P8 — Guild feedback request

"The way how you should think about creating clothing for creatures with tails, horns, wings and so on. I didn't put my thought into it before but now it makes so much sense that they would need something specifically tailored for their needs."

C.4 Findings & discussion 4.4

P7 — Guild feedback request

"Smart accessories were another topic you mentioned in the last session which I haven't considered before. I think it would be somewhat difficult to achieve and most would be done in VFX, but it is a nice concept and I keep thinking on what I could do about it or if I should take into consideration a special setup for it"

P7 — Hands-on workshop personal response sheet

"Considerations about how the material should bend and shape for rigging depending on the fabric for realistic deformations."

P7 — Hands-on workshop personal response sheet

"Simulations (cloth) when rigging will depend a lot more on the material than before."

P2 — Guild 2, Medieval and Renaissance

"The explanations about construction, how people, you know, in a time when people had to make their own, make all of their own clothes, or at least all clothes had to be handmade. really helps me make more informed decisions in regards to like how these clothes would be handmade for these mutated people as well, even if it isn't mediaeval or Renaissance. Because in like off the top of my head would kind of, for example, like I would think of like, oh yeah, like stitching would be a way to showcase that. But the kind of stitching and its functions, it's not something that I'm very aware of or what they do."

P2 — Guild 2, Medieval and Renaissance

"There is information you're getting that is kind of subconscious. Like even if I don't realise that purple is a rich colour. I might expect rich people when I see something in purple"

P2 — Guild 1, Introduction and foundations

"You know, in a world where no one really has the same body shape and like, like radically even more so than in real life, it's, you know, it's important to know how clothes are made." - P2

P1 — Guild 1, Introduction and foundations

"I wanted to make it sort of historical, believable, maybe not that accurate, but believable so that it could look like if it was worn back then I did look up for like real patterns from the pieces of garment I wanted to make."

P1 — Guild 1, Introduction and foundations

"I also focused on textiles so that I used the correct visual representation of the textiles, which I did pretty well I think for the first time."

P10 — Hands-on workshop personal response sheet

"How paper patterns work when you connect them to make a 3D shape"

P1 — Guild 1, Introduction and foundations

"also when doing the simulation in Marvelous, it helped. It's getting the folds on the sort of right place because I had to do not a lot of extra sculpting to make the outfit believable enough that it was like realism to semi realism."

P4 — Guild 4, ancient to folk and cultures

"I was really surprised by the creature adapted clothing because I have never really thought too much in depth how the clothing would need to be adapted or changed for the creature to actually be able to wear it. And honestly, I really thought that you can just like... Cut the hole in there for the wings, for example, or something like that, but when I was thinking about it in the morning, it just didn't make sense, and now I got the answer, so that was really nice to know."

P2 — Guild 2, Medieval and Renaissance

"There is information you're getting that is kind of subconscious. Like even if I don't realise that purple is a rich color. I might expect rich people when I see something in purple"

P5 — Guild 4, ancient to folk and cultures

"I also like the irony of it and I how much it can tell you like the society with like the more like supposedly useful you are, the less useful you dress actually"

P4 — Guild 4, ancient to folk and cultures

"It was also really nice to actually specify the hierarchy like why people in higher positions wore more layers and more complex clothing, and the opposite for lower hierarchy. It was nice to understand the reasons behind it."

C.5 Findings & discussion 4.5

P2 — Guild feedback request

"Going through the history of clothing helps you understand how things came to be the way they are now."

P1 — Guild feedback request

"New interesting topics and ages being explored each Guild session kept me interested and made me don't want to miss out on the new information."

P7 — Guild feedback request

"I enjoyed having the history presented. It is my preferred way of receiving information actually, knowing the why and the how."

P8 — Guild feedback request

"I really enjoyed the structure where you explain some kind of topic and then you have a question for Attendees."

P8 — Hands-on workshop personal response sheet

"The workshop you organised was well structured."

P8 — Hands-on workshop personal response sheet

"It was very fun and chill. The group was very nice, so it felt like we were collaborating at all times, which made experimenting more fun and less overwhelming"

P7 — Hands-on workshop personal response sheet

"Fun & enjoyable. Especially liked that we could experience the materials ourselves . I haven't really considered the weave before."

P9 — Hands-on workshop personal response sheet

"Really liked the draping and fabric sliders. Will help translating my fabric knowledge to digital."

P2 — Hands-on workshop personal response sheet

"It was a lot of fun. The different materials were very cool and interesting. The folds and drapes were helpful for my future drawings and the patterns were fun."

P9 — Hands-on workshop personal response sheet

"I think having the exercise papers individual would have helped a bit with data collection. I sometimes struggled to find the convenient moment to write, so some thoughts I had did not make it to the paper."

P9 — Hands-on workshop personal response sheet

"Working in a group I liked maybe less."

P7 — Guild feedback request

"I would like to see a few more in person workshops or maybe weekly challenges (though I don't know how they would be received). I think it would be cool to have people design something before a session, something simple, and then discuss difficulties and the process for a few minutes."

P1 — Guild feedback request

"I would like a focus on how to actually apply it into 3D characters for the student's projects. I think I would also have one or more Guild sessions where you break down a game character's garment."

P1 — Guild feedback request

"It would be cool if we / you could reconstruct clothes from game characters so we could learn about what to think about during the process and what to watch out for. Create a chosen garment together in the Guild and supply feedback and help"

P2 — Guild feedback request

"More like a walk-in style, chill discussion would be nice. Discussing people's work regarding clothing and costumes, making it like a group brainstorm session. We have gone over the theory now, but now we need to practice using that theory for our ideas and work."

P8 — Guild feedback request

"Specific environments or climates such as what people are thinking when creating designs for underwater environment in real life or with fantasy/creature characters. The underwater was the first idea but other could be for example around extreme conditions such as Freezing or Extremely warm temperatures and how those conditions affect clothes in terms of weathering."

BIBLIOGRAPHY

- Adams, J. F. (2023). *Teaching Costume Design and Costume Rendering: A Guide for Theatre and Performance Educators* (1st ed.). Routledge.
- Anthropic. (2026). *Claude*. In (Version Various versions) <https://claude.ai>
- Autodesk. (n.d.). *Maya*. In [Computer software]. <https://www.autodesk.com/products/maya>
- Blender Foundation. (n.d.). *Blender*. In [Computer software].
- Braun, V., & Clarke, V. (2006). Using thematic analysis in psychology. *Qualitative Research in Psychology*, 3(2), 77-101. <https://doi.org/10.1191/1478088706qp063oa>
- Chavez, J. (2025). *The Complete AAA VIDEO GAME Character Art Pipeline* [Video]. YouTube. https://www.youtube.com/watch?v=x7l4_jjZy-c
- CLO Virtual Fashion. (2024). Marvelous Designer wins 2024 Academy Award. <https://www.clo3d.com/en/resources/notices/494>
- CLO Virtual Fashion. (n.d.). *Marvelous Designer*. In [Computer software].
- Cork, J. (2015). Haute Future: How Fashion Designers Improved Deus Ex. <https://gameinformer.com/b/features/archive/2015/04/24/haute-future-how-fashion-designers-improved-adam-jensen-s-deus-ex-coat.aspx>
- Czauderna, A. (2018). Academic Game Design Education: A Comparative Perspective. In *Serious Games* (pp. 9-12). https://doi.org/10.1007/978-3-030-02762-9_2
- Davies, P. (2017). *The Art of Horizon Zero Dawn*. Titan Books.
- Eberle, H., Gonser, E., Hermeling, H., Hornberger, M., Kupke, R., Menzer, D., Moll, A., Kilgus, R., & Ring, W. (2014). *Clothing technology ... from fibre to fashion* (Sixth English edition ed.). Verlag Europa-Lehrmittel, Nourney, Vollmer.
- Epic Games. (2021). Balenciaga Brings High Fashion to Fortnite. <https://www.epicgames.com/site/en-US/news/balenciaga-brings-high-fashion-to-fortnite>
- Fashion Institute of Technology NYC. (n.d.). *Fashion History Timeline*. <https://fashionhistory.fitnyc.edu/>
- Franz, N. K. (2014). *The Unfocused Focus Group: Benefit or Bane?* (Vol. 16). <https://doi.org/10.46743/2160-3715/2011.1304>
- Gear Solid. (2024). *Marvelous Designer vs Blender vs Clo3D vs Maya vs Houdini: Cloth simulation Showdown 2025!* [Video]. YouTube. <https://www.youtube.com/watch?v=bxpNCUZ8lrs>
- Hawes, H. (2021). Review – It Takes Two. <https://waytoomany.games/2021/04/07/review-it-takes-two/>
- Hazelight Studios. (2021). *It Takes Two*. In [Video game]. Electronic Arts.
- Honeycutt, L. D. (2021). A semiotic approach to visual analysis of dress: symbolic communication of clothing color, cut, and composition through the French film costumes of Anaïs Romand. https://doi.org/10.31390/gradschool_theses.5408
- InspirationTuts. (2022). *3D scanning For Video Games* [Video]. YouTube. <https://www.youtube.com/watch?v=BLLAEM4k8c4>
- InspirationTuts. (2025). *Best 3D Software For Sculpting in 2025* [Video]. YouTube. <https://www.youtube.com/watch?v=jxShvtbHOrg>

- Jablon-Roberts, S., & Sanders, E. (2018). A Theoretical Framework for the Creative Process of Theatrical Costume Design for Historically Set Productions. *Clothing and Textiles Research Journal*, 37(1), 35-50. <https://doi.org/10.1177/0887302x18796320>
- Karimfazli, P. (2019). Game Character Production: Industry Standards. *80.lv*. <https://80.lv/articles/game-character-production-industry-standards>
- Kolb, D. A. (1984). *Experiential learning: Experience as the source of learning and development*. Prentice Hall.
- Kucharski, J. (n.d.). On The Upswing: Costume Design for Video Games. <https://tyrannyofstyle.com/costume-design-for-video-games/>
- Lyons, S. A. (2020). *Costume Design for Video Games: An Exploration of Historical and Fantastical Skins*. CRC Press.
- Makryniotis, T. (2018). Fashion and Costume Design in Electronic Entertainment—Bridging the Gap between Character and Fashion Design. *Fashion Practice*, 10(1), 99-118. <https://doi.org/10.1080/17569370.2017.1412595>
- Marvelous Designer. (2017a). Cyberpunk 2077. <https://www.marvelousdesigner.com/userspotlight?keyword=Game&postId=a7f9011def9f4c85bdc073da221a9146>
- Marvelous Designer. (2017b). The Witcher 3: Wild Hunt. <https://www.marvelousdesigner.com/en/userspotlight/post/987b23e0fa23402a84be5f8f9436d86b>
- Maxon. (n.d.). ZBrush. In [Computer software]. Maxon. <https://www.maxon.net/zbrush>
- Merriam, S. B. (2009). *Qualitative Research A Guide to Design and Implementation*. Jossey-Bass.
- Microsoft. (2025). *Teams*. In [Computer software]. Microsoft. www.microsoft.com
- Miro. (2026). *Miro Visual Workspace*. In [Computer software]. www.miro.com
- Neonakis, A. (2022). Costuming a Character's Narrative Arc. Lightbox Expo, Pasadena, California, USA.
- OBS Project Contributors. (2025). *Open Broadcasting Software*. In (Version 30.0) [Computer software]. <https://www.obsproject.org/>
- Paine, R., & McMahon, B. (2023). *The workshop model: Teaching ways of knowing and doing* LearnXDesign 2023,
- Patton, M. Q. (2002). *Qualitative research & evaluation methods* (3rd ed. ed.).
- Pikat. (2024). *How professional CONCEPT ARTISTS design characters* [Video]. YouTube. https://www.youtube.com/watch?v=9l_nuMGphW0
- Proko, & Evstratiy, E. (2024). *Character Design Secrets Everyone Should Know* [Video]. YouTube.
- Proko, & Zhang, K. (2021). *Level Up Your Character Design with Knight Zhang* [Video]. YouTube. <https://youtu.be/CjiKLE3NuI>
- Reddit user: FreeWear2215. (2025). Ellie Williams - The Last of Us II. In: Reddit.
- Riot Games. (2018). *So You Wanna Make Games?? / Episode 2: Concept Art* [Video]. YouTube. <https://youtu.be/FqX-UMVTLHI>
- Robinson, R. B., & Alvarez, A. (2024). *Toward a Design and Play-Focused Approach to Teaching Technical Game Design* Proceedings of the 19th International Conference on the Foundations of Digital Games,

- Robson, C., & McCartan, K. (2016). *Real World Research* (4th ed.). Wiley.
- Salomaa, H. (2018). Video Games and Costume Art -digitalizing analogue methods of costume design.
- Salomaa, H. (2019). Costume in Games: Integrating a Costume Designer into the Character Pipeline. Game developers conference, San Francisco, California, USA.
- Salomaa, H. (2021). Fashion or Costume? Defining the design approach for game character's style. Exthereal Conference, Online.
- Shahbazi, N. (2025). Realistic vs. Stylized Art in Games: Which Style Fits Your Vision? <https://pixune.com/blog/stylized-vs-realistic/>
- Stutesman, D. (n.d.). International history of costume design in film making. <http://www.filmreference.com/encyclopedia/Academy-Awards-Crime-Films/Costume-INTERNATIONAL-HISTORY-OF-COSTUME-DESIGN.html>
- Tepe, J., & Koohnavard, S. (2022). Fashion and game design as hybrid practices: approaches in education to creating fashion-related experiences in digital worlds. *International Journal of Fashion Design, Technology and Education*. <https://doi.org/10.1080/17543266.2022.2103591>
- Tereshchuk, A. (2025). Realistic vs Stylized Art in Video Game Development. <https://gamestudio.n-ix.com/realistic-vs-stylized-art-in-video-game-development/>
- The Metropolitan Museum of Art. (n.d.). *The Met Collection*. <https://www.metmuseum.org/art/collection>
- Tran, V. (2019). Fashion in Games: Why It Matters 2. <https://www.gamedeveloper.com/design/fashion-in-games-why-it-matters>
- vhv.rs. (2019). Jill Valentine - Resident Evil 3. In.
- Victoria & Albert Museum. (n.d.). *V&A Collections*. <https://www.vam.ac.uk/collections>
- Wade, D. (2021). Hazelight Studios It Takes Two Art Blast. <https://magazine.artstation.com/2021/04/hazelight-studios-it-takes-two-art-blast/>
- Zealous XR. (2025). CLO3D vs Marvelous Designer: Which Fashion Design Software is Right for You? <https://www.zealousxr.com/blog/clo3d-vs-marvelous-designer#id5>

